Lab Book

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Preface

This book details my research during Summer 2026. Many thanks to Dr. Andrew Young and Mr Darius Michienzi for supervising this project.

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1 Reading

1.1 Summary of "Broad Iron Lines in Active Galactic Nuclei" [1]

Fluorescent iron line, Fe-Kα, in the X-ray band at $6.4 - 6.9$ keV is the strongest narrow line emitted by an accretion disk around a black hole. Inverse Compton scattering of the soft photons (lower energy than the reactant particles) in a hot corona is the most likely candidate for the production of these X-rays. The emitted line is broadened and skewed due to the Doppler effect and gravitational redshift, and is often called the broad iron line due to being broadened to a FWHM of 5×10^4 km s^{-1}.

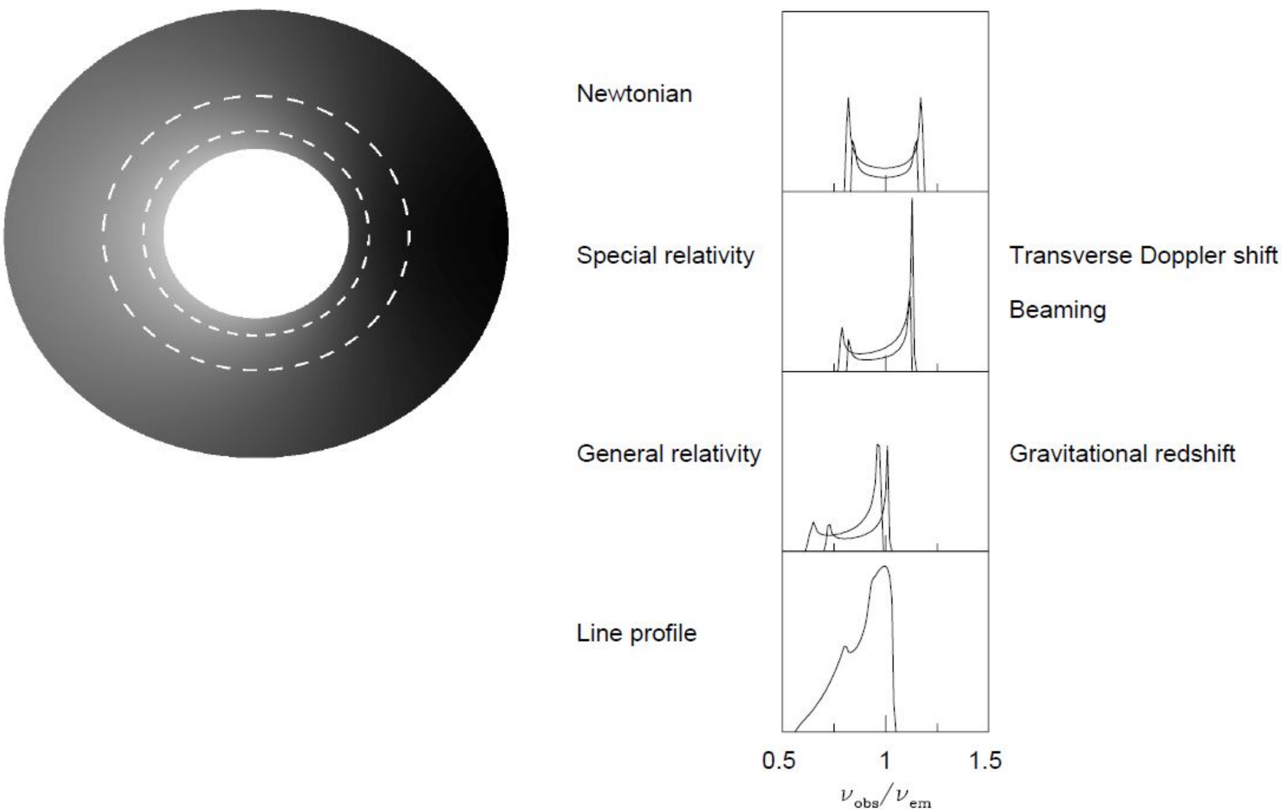


Figure 1.1: The profile of the broad iron line, image credit to Fabian (2000) [1].

Soft Comptonisation (multiple inverse Compton scattering by hot thermal electrons) occurs giving a power law X-ray spectrum. Flares from active or flaring regions irradiate the accretion disk forming a "reflection" component in the spectrum.

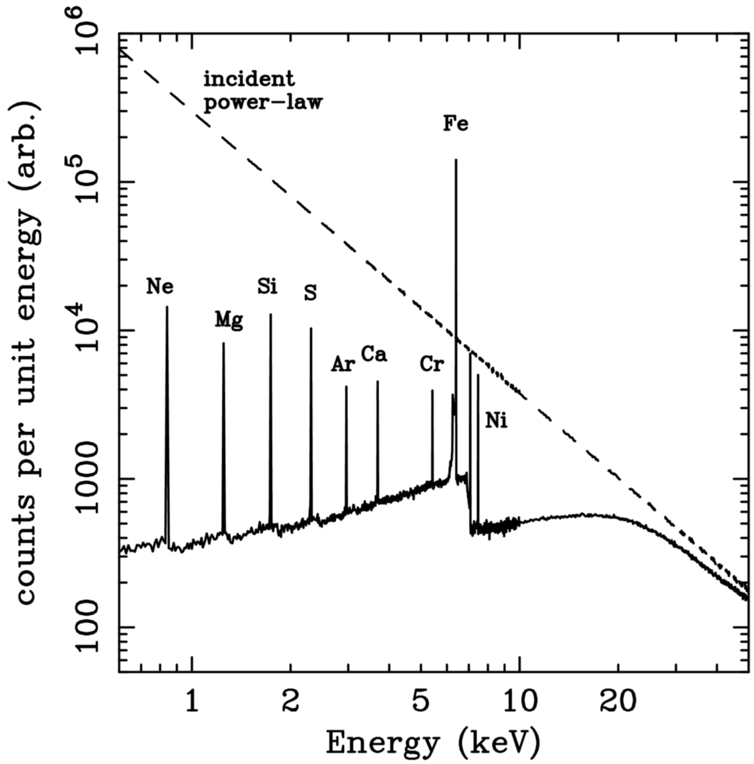


Figure 1.2: Reflection spectrum from a Monte Carlo simulation of an illuminated slab. The dashed line is the incident continuum and the solid line shows the reflected spectrum. Image credit Reynolds (1996) [1]

The iron line is produced when one of the two K-shell electrons is ejected following absorption of an X-ray, the threshold for neutral iron is 7.1 keV. The resultant state can decay through the dropping of an L-shell electron into the K-shell, releasing 6.4 keV as either an emission line photon (34%) or an Auger electron (66%). Ionised iron has a photoelectric threshold and the Kα of higher energy. The strength of the iron line is measured in terms of its equivalent width.

Equivalent Width

The equivalent width of a spectral line is the width of the rectangle which has the same area as that between the line and the continuum level.

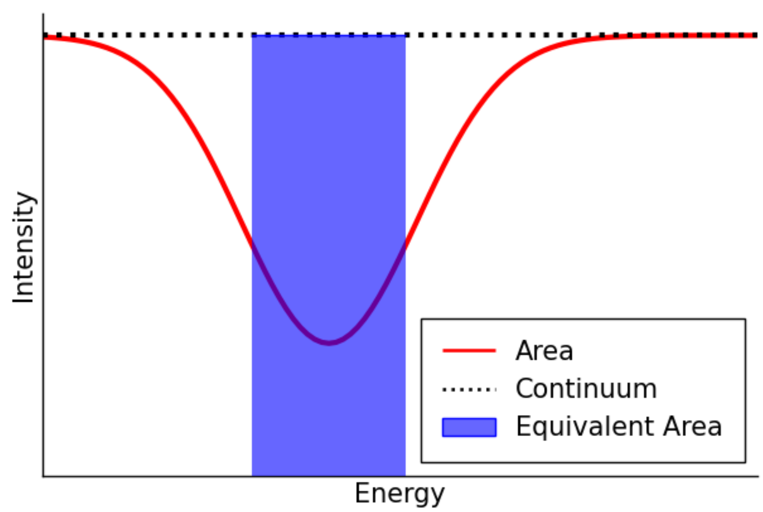


Figure 1.3: A diagram indicating the equivalent width. The width of the rectangle is the equivalent width.

The equivalent width is a function of the geometry of the accretion disk, the elemental abundances, the inclination angle, and the ionisation state.

! Key Points

- **X-ray Continuum**: The hard X-ray continuum in AGNs and GBHCs is thought to be from active/flaring regions in the corona. Thermal electrons multiply inverse Compton scatter optical and UV photons from the disk to X-ray energies. A hard X-ray power law results and irradiates the accretion disk to produce a reflection component.
- **Reflection Component**: The reflection component causes the spectrum to flatten above 10 keV as Compton recoil reduces the backscattered flux and results in a strong fluorescent iron line at ~ 6.4 keV. The energy and strength of this line depends on iron abundance, inclination of the disk, and its ionisation state. A strong iron absorption edge may also be present from an ionised disk.
- **Iron Line Profile**: Doppler shifts, relativistic broadening and gravitational redshifting give the broad and skewed line profile. As the line comes from the innermost regions, these effects are very pronounced and may be used to determine the black hole spin and inclination angle of the accretion disk. In most Seyfert 1 galaxies, the disk is inclined at around 30°, consistent with the standard model.
- **Observations**: The iron line was first clearly detected by Ginga and a line profile subsequently resolved by ASCA confirmed the broad and skewed shape expected from an accretion disk around a Schwarzschild black hole. RXTE has been used to study the line and continuum variability on shorter timescales, with reduced energy resolution.
- **Black Hole Spin**: The radius of the ISCO decreases with spin. Since the line profile is sensitive to the innermost radius of fluorescent emission this may be used to estimate the black hole, with some astrophysical assumptions. With present time-averaged observations such measurements may be ambiguous as alternative models with different values for the spin may produce identical line profiles.
- **Alternative Models for the Production of the Broad Iron Line**: Alternative models that do not require an accretion disk appear to fail, in particular the line width cannot be entirely due to Comptonisation. Hybrid models in which the Comptonisation and Doppler/gravitational effects produce the profile are heavily constrained.
- **Variability**: Rapid X-ray continuum variability is observed in most AGNs, and the iron line is expected to vary in response to this with short time lag. Whilst these timescales are too short to be probed presently, significant and complex iron line variability has been observed. Often the line flux is seen to remain constant whilst the continuum changes, and there appears to be an anticorrelation between the reflected fraction and the equivalent width. In another study the reflected fraction and the photon index of the power law are correlated for individual objects and between different ones. These need to be explained as they appear contrary to our simple model of reflection. Flux correlated changes in the ionisation state of the disk may explain some of these facts.
- **Reverberation Mapping**: The rapid variability is associated with activation of new flares in the corona above the accretion disk. X-ray reverberation mapping uses observations of the iron line response to sudden changes in the continuum to study the accretion disk and black hole. This may be used to determine the geometry of the X-ray emission and the black hole spin and mass.
- **Other Classes of Object** Iron lines are observed in quasars, LLAGNs, radio-loud AGNs, and GBHCs. Quasars have the iron line decrease with increasing luminosity which may be due to the more luminous sources accreting closer to the Eddington limit being more highly ionised. Observations in LLAGNs may determine whether the accretion with low rates is an advection-dominated accretion flow or a thin disk. Weak lines have been observed suggesting that their low accretion rate flows are thin disks rather than optically thick ADAFs. THe broad iron lines and reflection humps are weak or absent perhaps because the reflection signature is swamped by a beamed continuum. GBHCs show smeared absorption and emission features in the reflection spectra, and there is a debate as to the precise nature of the accretion flow within a few tens of Schwarzschild radii.

2 Initial Gradus Testing

The below details initial experimentation with `Gradus.jl`. The code block below renders an image of a black hole with a thick accretion disk. Colour denotes the redshift of a given region on the disk.

```
using Gradus
using Plots

# Function defining the cross section of the disk
function discCrossSection(x)
    centre = 8
    radius = 3

    if (x < centre - radius) || (radius + centre < x)
        zero(x)
    else
        r = x - centre
        0.5(sqrt(radius^2 - r^2) + (0.5sin(3x)))
    end
end

# Metric Parameters
M = 1.0     # Mass
a = 0.7     # Black hole spin
 13 = 0.0
 22 = 0.0
 52 = 0.0
 3 = 0.0
i = deg2rad(70) # Inclination

# Instantiating the metric
m = JohannsenMetric(M, a,  13,  22,  52,  3)

# Position of the observer
# (x[1], Distance, Inclination, x[4])
x = SVector(0.0, 1000.0, i, 0.0)

# Maximum affine time, parameterises the solution
#  _max ~ 2*x[2]
 _max = 2000.0

# Setting up impact parameters
  = range(-10.0, 10.0, 100)
  = range(-10.0, 10.0, 100)
```

```
# Getting an array of velocities from the impact parameters
vs = vec([map_impact_parameters(m, x, a, b) for a in  , b in  ])
xs = fill(x, size(vs))

# Generating solutions
sols = tracegeodesics(m, xs, vs,  _max, save_on = false)

# Generating points and reshaping to be the same shape as the image
points = unpack_solution.(sols.u)
points = reshape(points, (100, 100))

# Redshift point function
redshift = ConstPointFunctions.redshift(m, x)
redshiftGeometry = redshift   ConstPointFunctions.filter_intersected()

# Disc
d = ThickDisc(r -> discCrossSection(r))

# this function returns the impact parameter axes
 ,  , image = rendergeodesics(
    m, x, d,  _max, pf = redshiftGeometry,
    # image parameters
    image_width = 200, image_height = 150,
    # the "zoom" -- use the impact parameter axes
     lims = (-20, 20),  lims = (-15, 15), verbose = false
)

heatmap( ,  , image, aspect_ratio = 1)
```

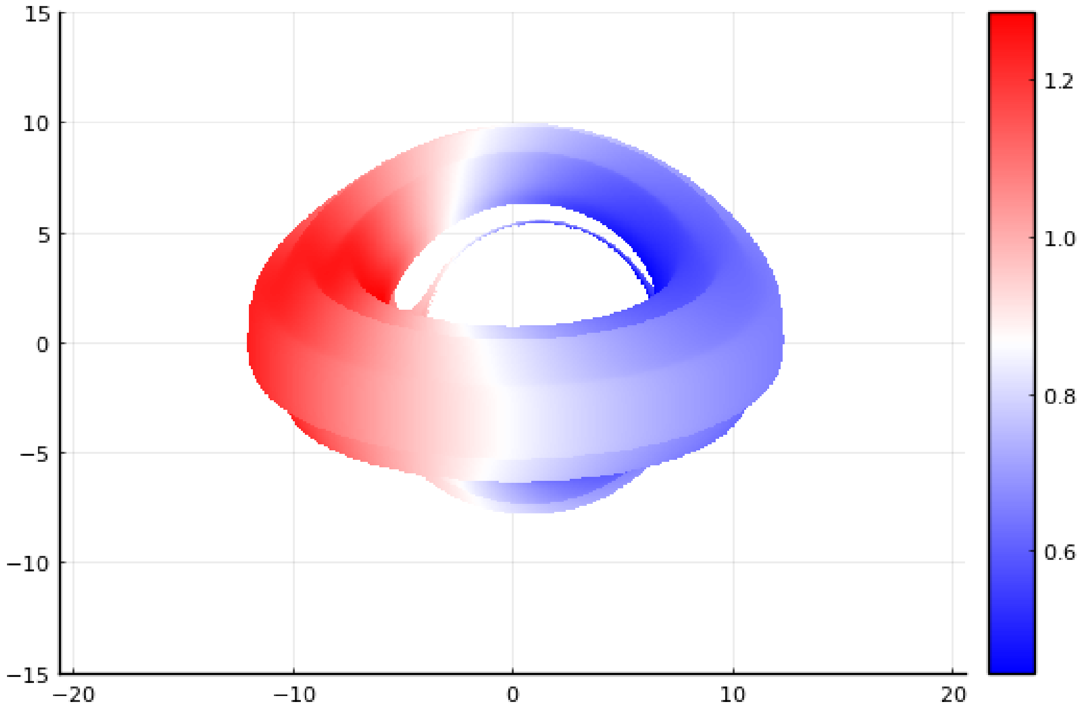


Figure 2.1: The first render produced.

Once familiarity was gained with the basics of `Gradus.jl` the line profile functionality was utilised through the following function.

```
function computeLineProfile(m, i, height, bins = range(0.0, 1.5, 180))
    # Position of the observer
    # (x[1], Distance, Inclination, x[4])
    x = SVector(0.0, 10000.0, i, 0.0)

    # Disk
    d = ThinDisc(0.0, Inf)

    # Setting up the model and profile
    model = LampPostModel(h = height)
    profile = emissivity_profile(m, d, model)

    # Computing the line profile
    bins, flux = lineprofile(m, x, d, profile; bins=bins, verbose=true)

    return bins, flux
end
```

This will allow for the investigation of the free parameters for the Johannsen metric; α_{13}, α_{22}, α_{52}, and ϵ_3. A lamppost corona model is used with variable height `h`. An example of the effect of varying the height of this corona is shown below.

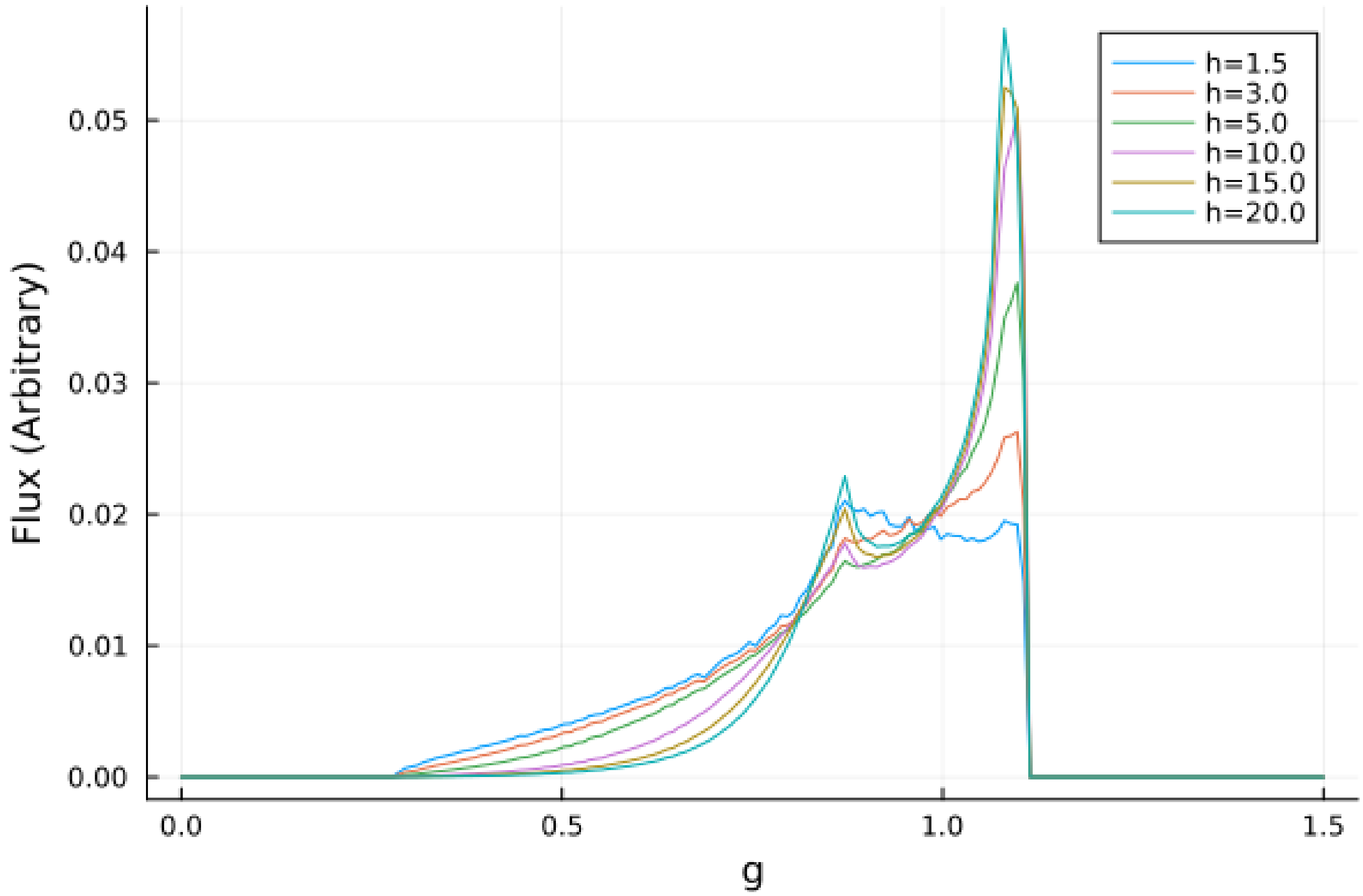
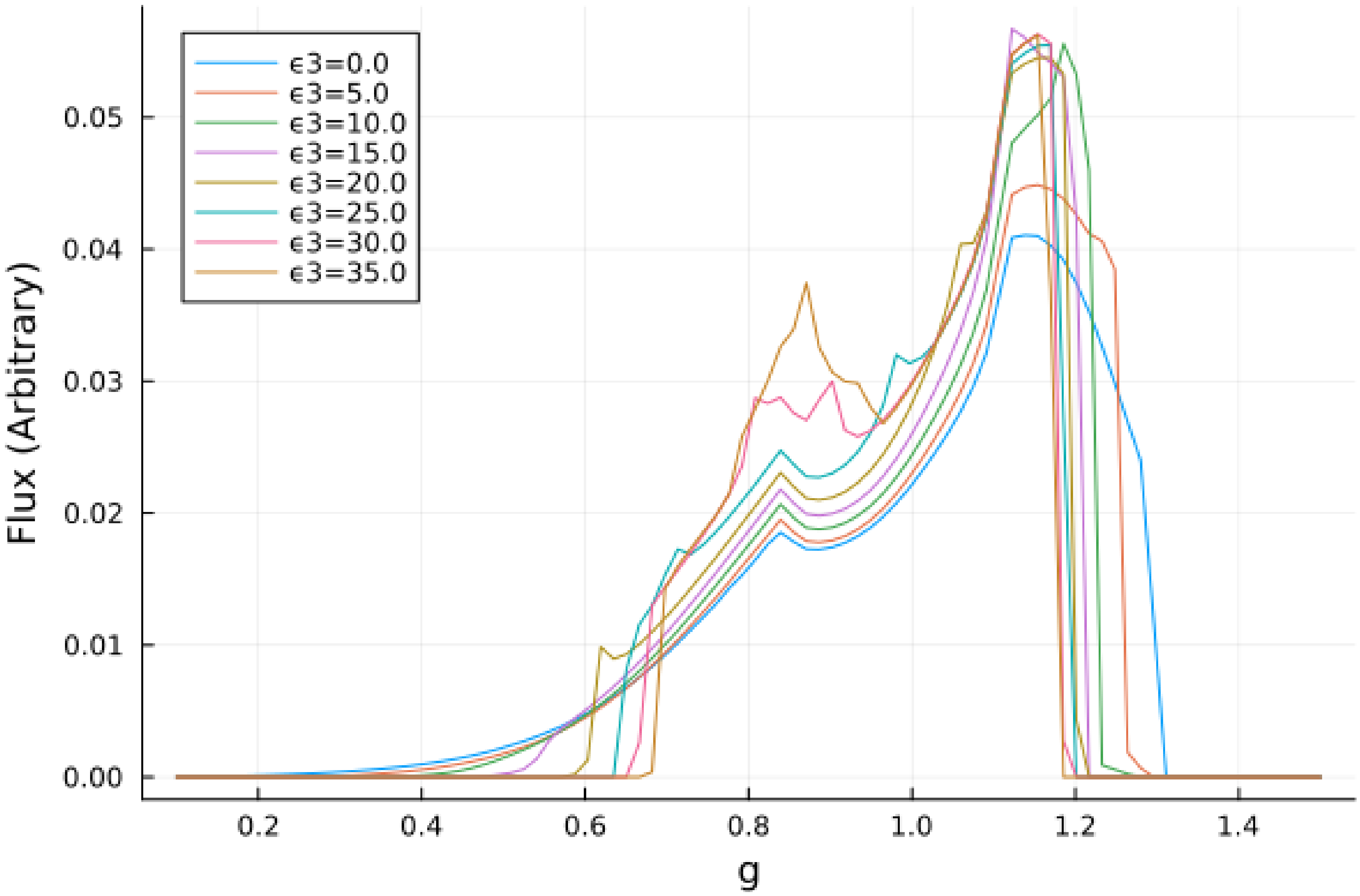


Figure 2.2: Initial variations of the height of the lamppost corona.

Following initial experimentation the two functions above were integrated together for efficient investigation. The two free parameters that will be the focus of this investigation are α_{13} and ϵ_3. Examples of the effects of these parameters on the emission lines are shown below


ϵ3=0.0
ϵ3=5.0
ϵ3=10.0
ϵ3=15.0
ϵ3=20.0
ϵ3=25.0
ϵ3=30.0
ϵ3=35.0
Flux (Arbitrary)
0.00
0.01
0.02
0.03
0.04
0.05
0.2
0.4
0.6
0.8
1.0
1.2
1.4
g

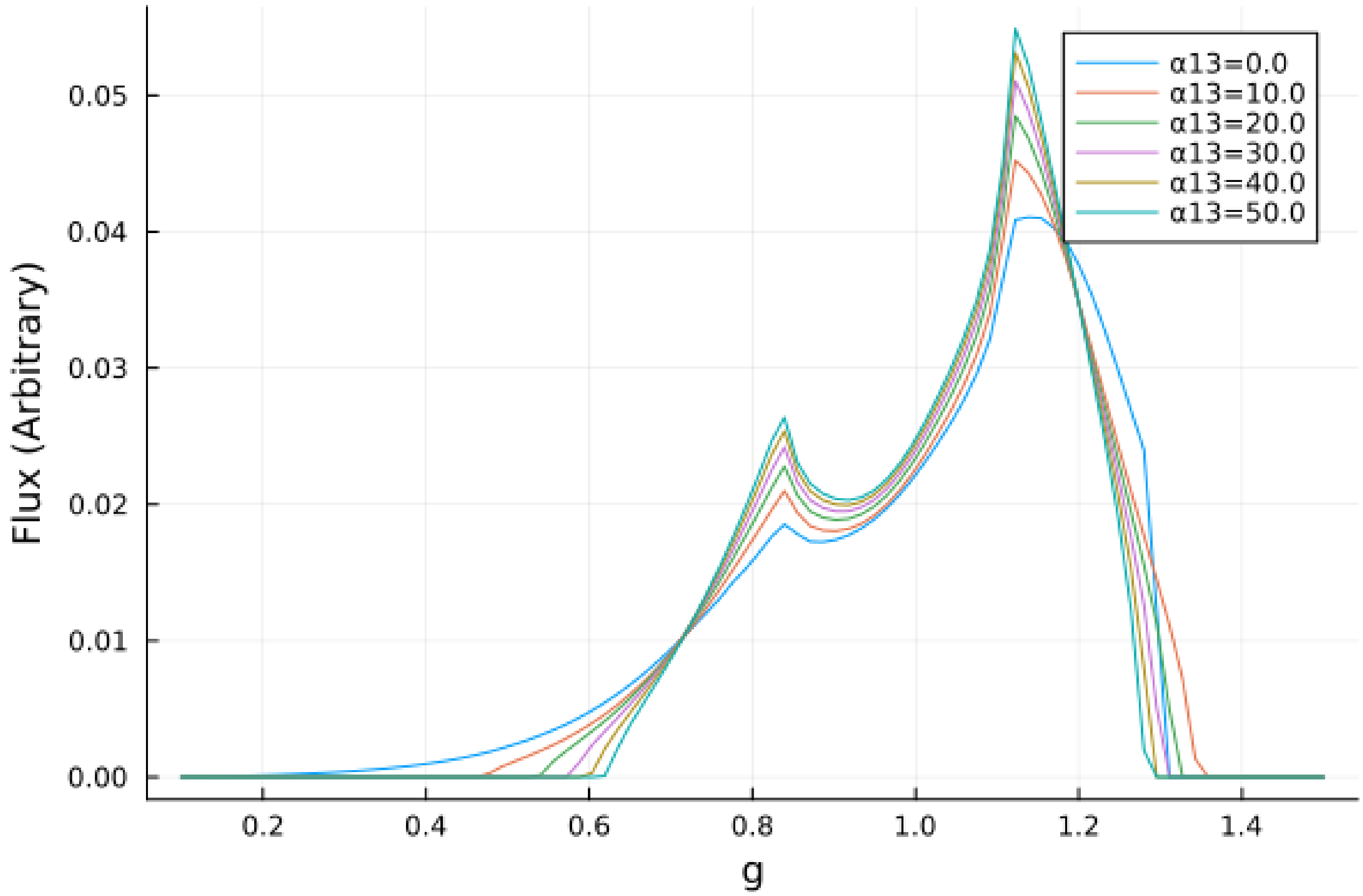


Figure 2.3: Initial variations of α_{13} and ϵ_3

It can be seen that α_{13} has a very smooth, almost linear effect on the shape of the line, while ϵ_3 is much more chaotic, particularly for $\epsilon_3 > 25$.

3 Parameter Space Investigations

Code was set up to easily run several variations to the model which will allow for the investigation of the parameter space. A thin disk accretion model and a lamppost corona were used. It was set up such that for a given configuration of h, a, i, ϵ_3, and α_{13} the line profile would be calculated and an image of the accretion disk produced and plotted together.

3.1 Ruling out α_{22} and α_{52}

Before exploring the 5D parameter space, tests were ran on both α_{22} and α_{52} such that their effects could be noted, should they be useful in the future.

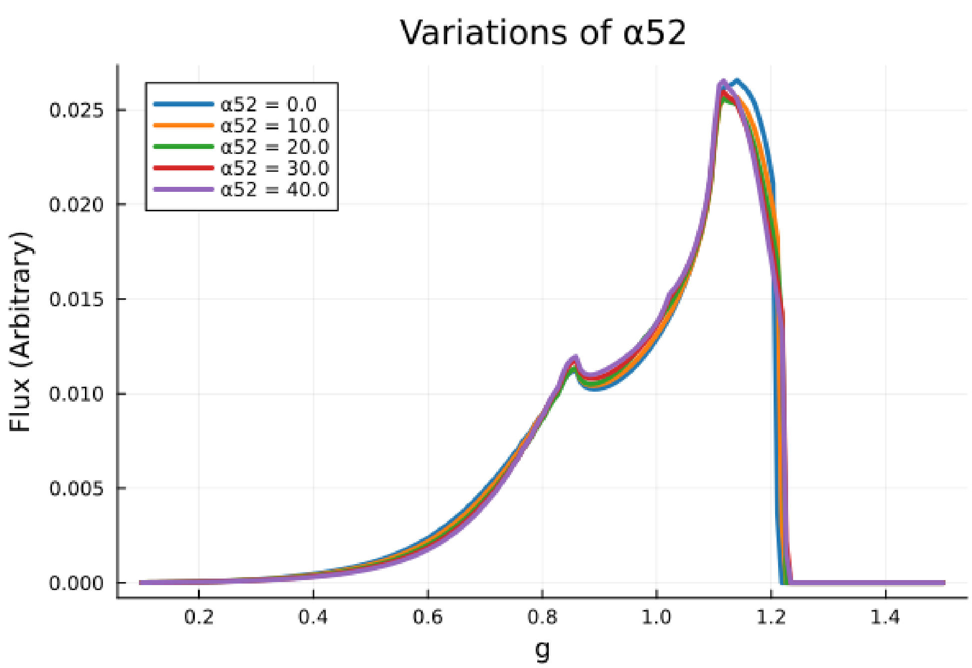


Figure 3.1: The resultant line profile from varying α_{52} between 0 and 40. A value of 50 did not successfully compute suggesting an upper limit on this parameter.

Varying α_{52} showed near random, and near zero effect on the line and will thus be disregarded for this investigation. Varying α_{22} often failed to compute and will thus similarly be disregarded for this investigation.

3.2 Exploring the 5-dimensional parameter space

Before varying the parameters it was necessary to determine the limits for useful variations. The spin is fundamentally limited to be $|a| < |M|$ and will thus be varied between 0 and 1. The ranges and default values for the parameter space are tabulated in Table 3.1.

Table 3.1: The ranges and default values for the parameter space.

Parameter	Minimum	Maximum	Default
α_{13}	0.0	50.0	0.0
ϵ_3	0.0	30.0	0.0
a	0.0	1.0	0.998
h	R_{ISCO}	30 R_G	10 R_G
i	0.0°	90.0°	60°

3.2.1 Line profiles

- Figure 3.2 shows that increasing the free parameter α_{13} sharpens the two peaks of the line, increases the maximum flux, and makes it more narrow.
- Figure 3.3 shows that increasing the free parameter ϵ_3 similarly increases the maximum flux, however the resultant profile is much more noisy. While α_{13} provided sharp peaks, ϵ_3 provided broader peaks, particularly in the shorter one.
- Figure 3.4 shows that increasing the inclination smoothes out the line profile and shifts the peak to higher energies. The line is broadened significantly.
- Figure 3.5 shows that increasing the corona height makes the line profile sharper, increases the maximum flux, and shifts the maximum to slightly lower energies.
- Figure 3.6 shows that the spin had the least prominent impact on the profile of the line, seeming to only slightly decrease the height of the peaks with increasing spin. This may be an effect of the small range of spins available. It may therefore be informative to investigate this further by increasing the mass, using different corona heights, or varying the inclination.

3.2.2 Redshift Image Renders

Alongside the line profiles, images of the accretion disk were rendered with a colourmap denoting the redshift of light from a given part of the disk. The corona height was not varied here as it did not impact the render.

Some of the parameters appeared to change the radius of the ISCO. This was investigated by explicitly finding the ISCO, as tabulated below.

Parameter	Value	ISCO (R_G)
α_{13}	0.00	1.24
	16.67	8.62
	33.33	11.70
	50.00	14.03
ϵ_3	0.00	1.24
	10.00	2.36
	20.00	2.90
	30.00	3.51
a	0.00	6.00
	0.32	4.92
	0.63	3.68
	0.95	1.94

The ISCO is proportional to both α_{13} and ϵ_3, and inversely proportional to the spin.

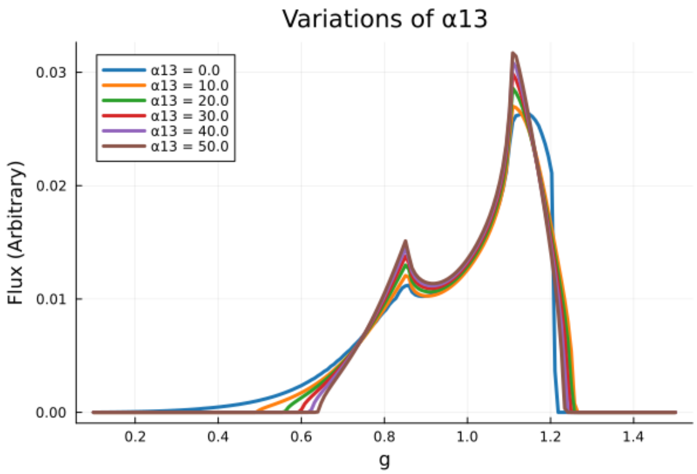


Figure 3.2: The line profiles resultant from varying α_{13} between 0 and 50.

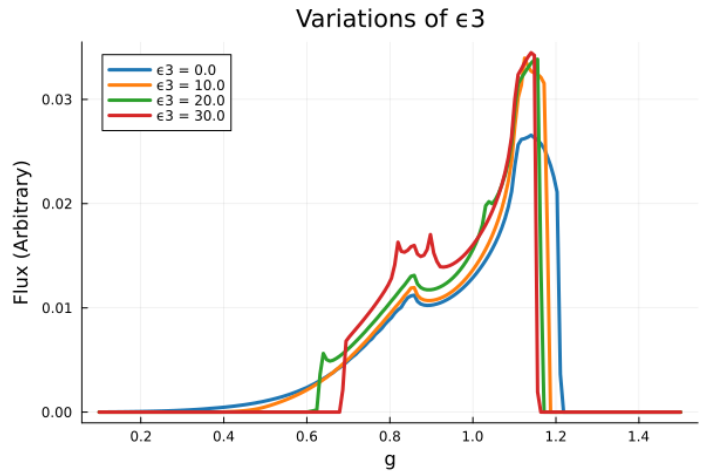


Figure 3.3: The line profiles resultant from varying ϵ_3 between 0 and 30. Values of 40 and 50 were tested however caused an error, there must be an upper bound on the allowed values of this parameter.

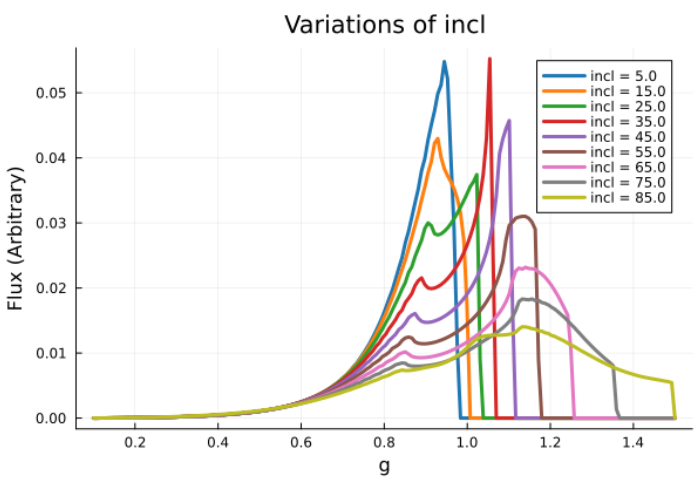


Figure 3.4: The line profiles resultant from varying the inclination between 5 and 85 degrees.

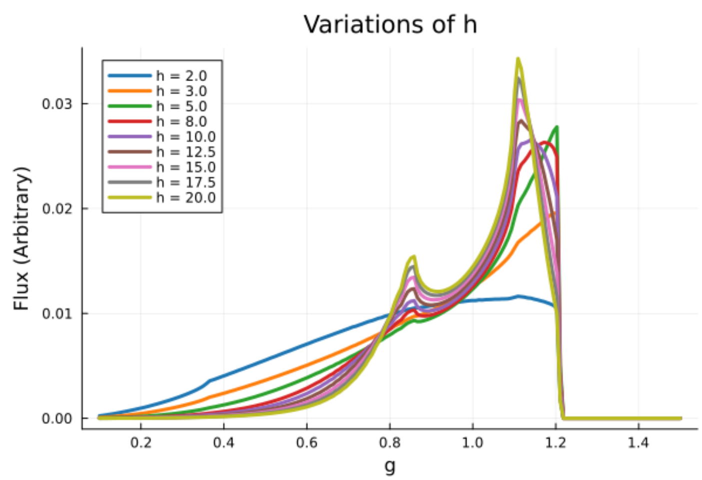


Figure 3.5: The line profiles resultant from varying h between 5 and 30 gravitational radii.

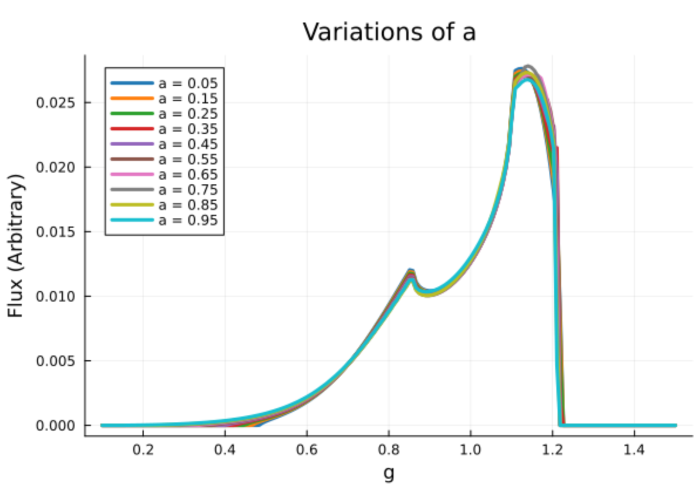


Figure 3.6: The line profiles resultant from varying a between 5 and 95% of $M(=1)$.

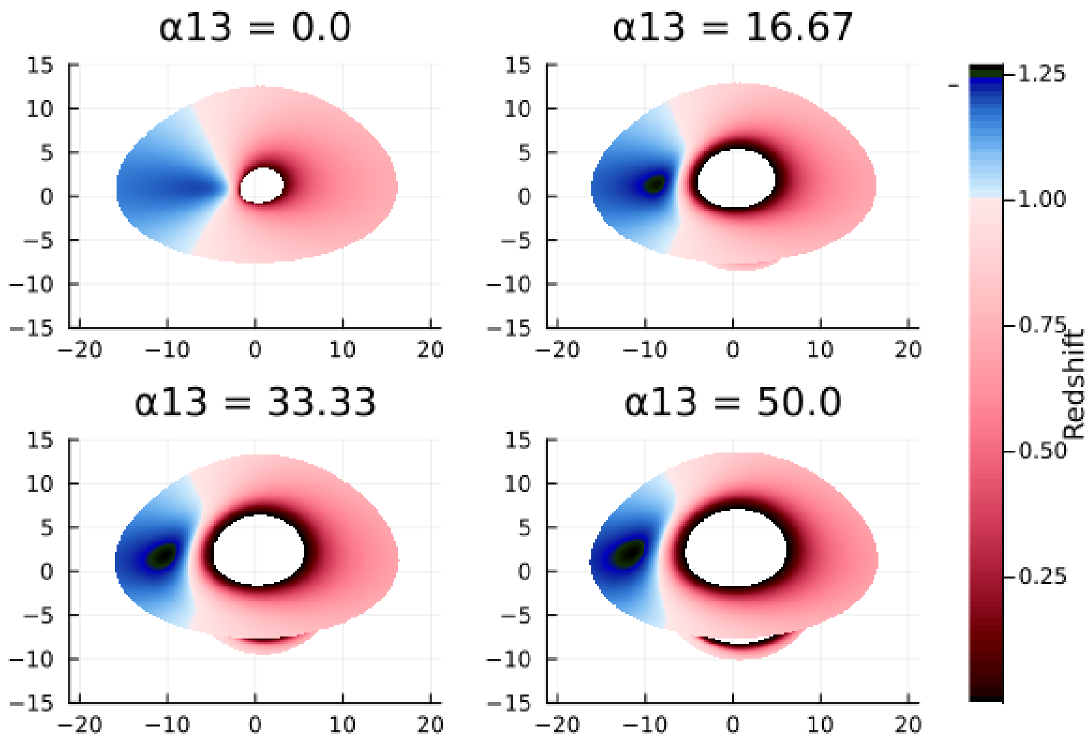


Figure 3.7: The effect of α_{13} on the accretion disk render. The ISCO appears to grow with α_{13}

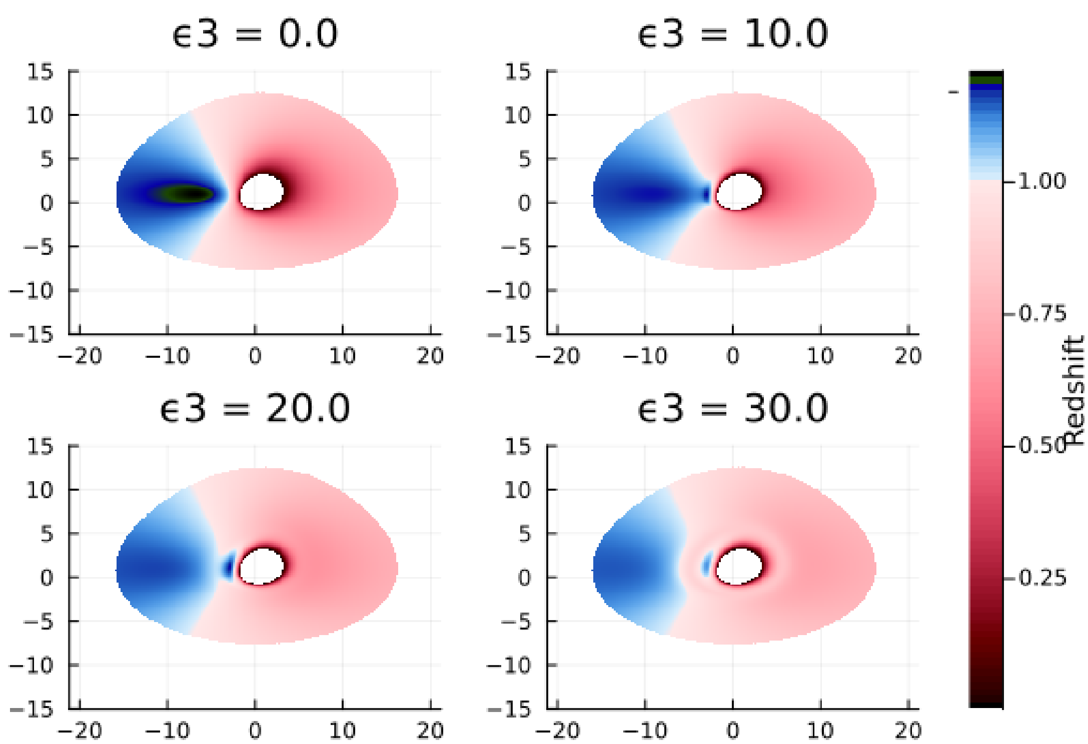


Figure 3.8: The effect of ϵ_3 on the redshift image. The shape of the disk does not change but the redshift profile does, particularly in the shrinking of the blueshift region, and an overall decrease in the degree of redshifting.

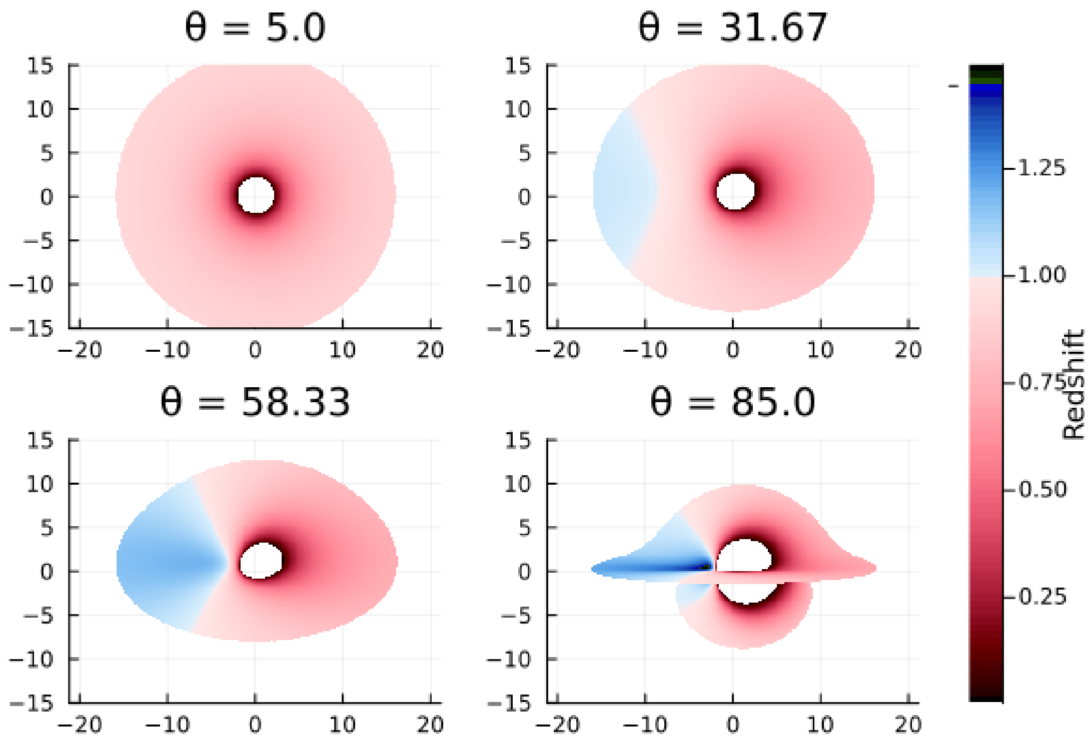


Figure 3.9: The effect of inclination on the redshift image. The redshift becomes more apparent as the inclination increases and the disk becomes edge on. This agrees with the observed line profile trend of energy increasing with inclination as the effects of doppler shifting become more apparent with increasing energy.

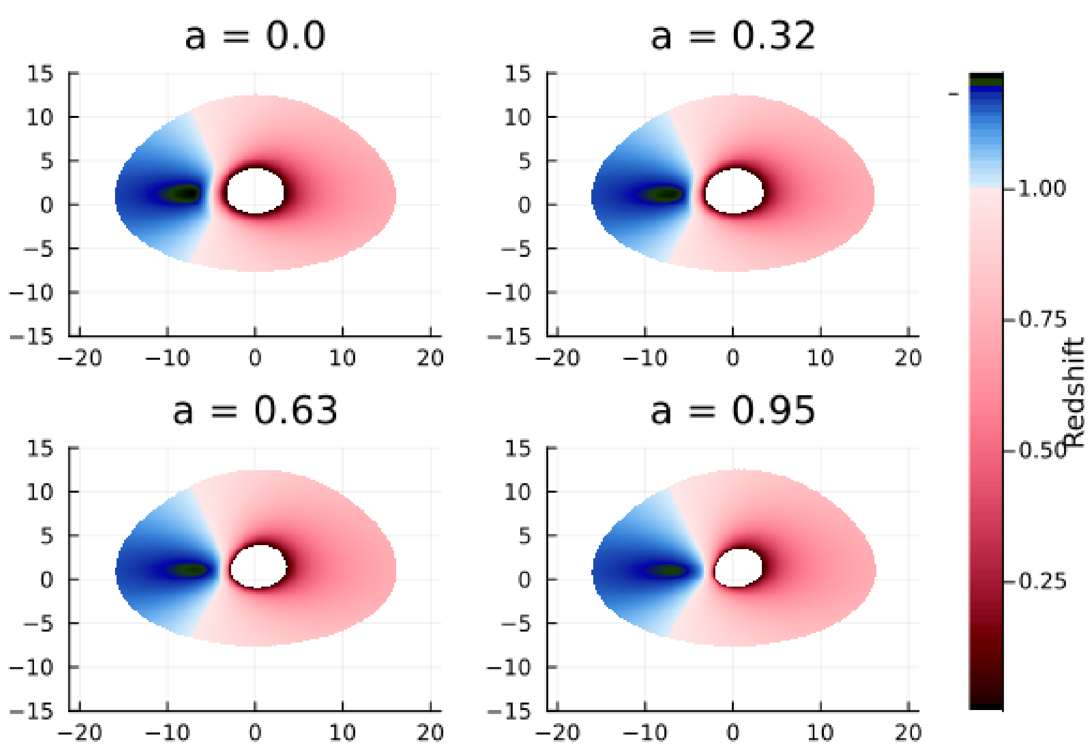


Figure 3.10: The effect of black hole spin on the redshift image. The ISCO appears to shrink.

3.3 Summary

> **!** Summary
>
> - α_{22} and α_{52} had unpredictable or minimal effect on the line profile and rendered images so are therefore to be disregarded.
> - a: Increasing the spin appears to shrink the ISCO of the accretion disk. Its effect on the line profile is less obvious, it appears to decrease the flux slightly but is otherwise negligible.
> - h: Increasing the corona height makes the line profile sharper and increases the peak flux, while decreasing the energy of the peak. It has no effect on the renders.
> - θ: Increasing the inclination (where 0 degrees has the accretion disk face-on) smoothes out the line profile and increases the peak energy. The effect of redshift becomes more apparent with increased inclination.
> - α_{13}: Increasing α_{13} appears to increase the radius of the ISCO and make the line profile sharper.
> - ϵ_3: Increasing ϵ_3 increases the height of the line profile, and makes it more chaotic.
> - The ISCO is proportional to both α_{13} and ϵ_3, and inversely proportional to the spin.

3.4 Code Specifics

The workhorse of the parameter investigations was the utility function `JohannsenParamVar` which sets up the metric and observer position and then calls either `ComputeLineProfile` or `RenderImage`. This utility function was made versatile using keyword arguments allowing for different methods to be called within it. This was then wrapped in `ParamLoop` which took the index (the variable to be looped through) and a set of values to call `JohannsenParamVar` for.

To maximise readablity of the redshift images, a custom colormap is made through `RedshiftColormap` which gives red for any values < 1, blue for those > 1, and white for those $= 1$. This gave the images seen in Figure 3.7, 3.8, 3.9, 3.10.

4 Fitting Spectral Lines

With the parameter space explored, it was useful to setup the architecture for fitting the model to spectral data. This was done using `SpectralFitting.jl` and defining a custom model using Gradus which fits the following parameters:

Parameter	Meaning
`K`	Normalisation
`h`	Corona Height
`E`	Line Energy
`R_in`	Inner Radius
`R_out`	Outer Radius
	Inclination
`a`	Spin
`13`	α_{13} Parameter
`3`	ϵ_3 Parameter

The architecture of this model was heavily based upon previous work on GitHub.

The model is instantiated using the `LampPostJohannsen` utility function. When called, the parameters are used to instantiate a `JohannsenMetric`, observer position, `ThinDisc`, and an emissivity profile. These are then used to compute the line profile as previously done which is returned.

All fitting is handled by `SpectralFitting` through the Levenberg Marquadt algorithm.

To test the implementation a line profile was computed and random noise added to it. This was then passed through the fitting function. The fitting process currently takes a long time and yields poor results so more work is needed. Time was taken to streamline the functionality of main2.jl via keyword arguments and general code cleanup.

Upon fitting, the error `Warning: No parameter uncertainty estimation due to error: LinearAlgebra.Singular` occurs and therefore the fit is not currently quantified.

With the current errors present it was decided to attempt the fit on actual data in order to determine whether they are a result of the data, or the fitting algorithm.

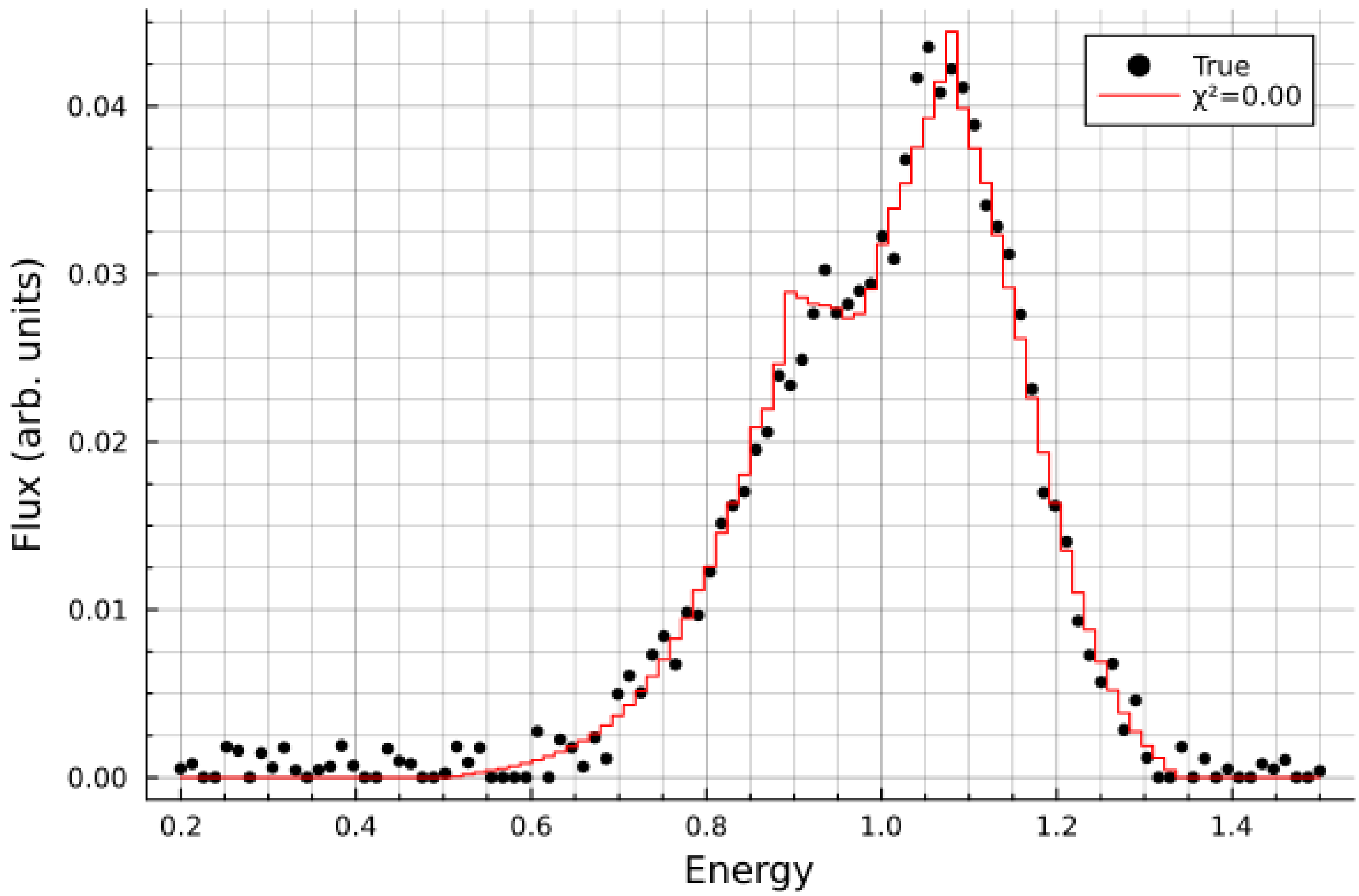


Figure 4.1: The first attempt at fitting to the noisy simulated data. This fit took 67 iterations and 15.3 minutes to complete.

5 Fitting to Data

Data from the NuSTAR (Nuclear Spectroscopic Telescope Array) spacecraft [2] was used, and the dataset used for testing was `nu80402315002B01`. Data loading was handled by `SpectralFitting.jl` through the `OGIPDataset` struct. This takes the spectrum file as an argument and automatically pulls in the background, `.arf`, and `.rmf` files to compute the spectrum.

5.1 Initial Results

Table 5.1: The ranges and default values for the parameter space.

Model	Variable	Value	Error	χ^2
Johannsen	a	0.57	1.88	116.84
	h	50.00	8.77	
	θ	58.24	4.93	
	α_{13}	2.29	10.96	
	ϵ_3	29.74	25.89	
	K	161.8	7.15	
	E	6.27	0.016	
Kerr	a	0.77	0.057	124.64
	h	50.0	2.75	
	θ	55.44	2.67	
	K	162.0	7.07	
	E	6.27	0.014	

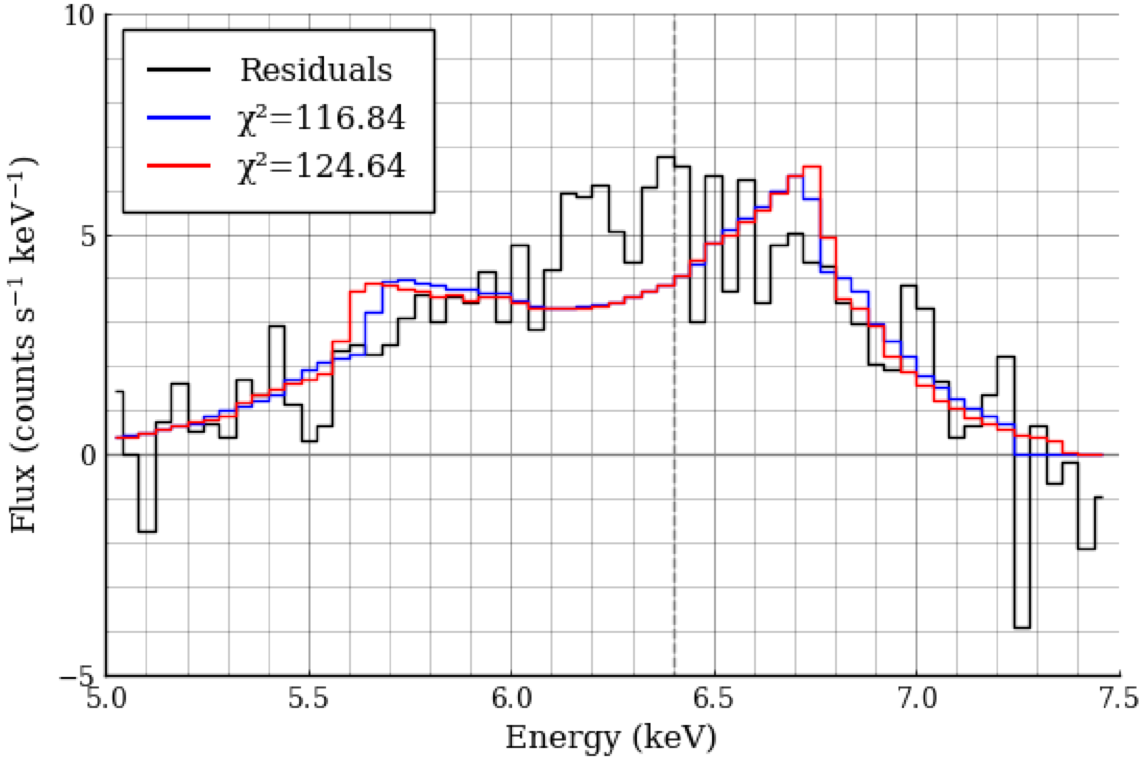


Figure 5.1: The first attempt at fitting to data. The Johannsen metric was used to produce the blue line, and the Kerr metric was used to produce the red. Neither fit was particularly good which may either be the result of poor data, or a poor model.

The NuSTAR datasets have two distinct spectra from each telescope, which can be fit simultaneously, binding some of the parameters together. The fitting code was thus adjusted to fit both datasets simultaneously while binding all parameters other than the normalisation. The dataset labelled "nu80402315002" was once again used.

It is evident from Figure 5.2 that the fitting process needs refinement as the fits do not currently fit the data well, and therefore drawing comparisons would be useless.

Following these experiments it was decided to fit a composite model of `PowerLaw` + `LampPost`, with the latter being the line profiles generated using either the Kerr or Johannsen metric through `Gradus.jl`. The first attempt at this provided the plots below.

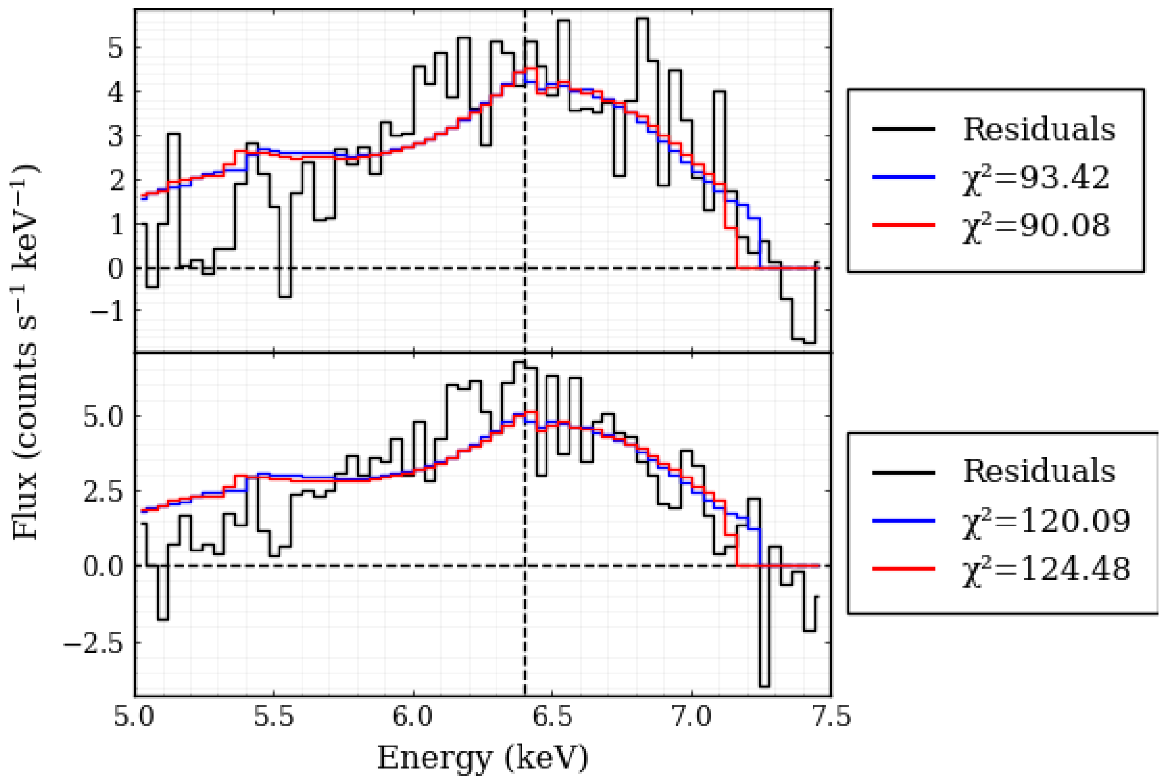


Figure 5.2: The fits to the nu80402315002 dataset. The Johannsen fit is shown in blue, and the Kerr fit in red.

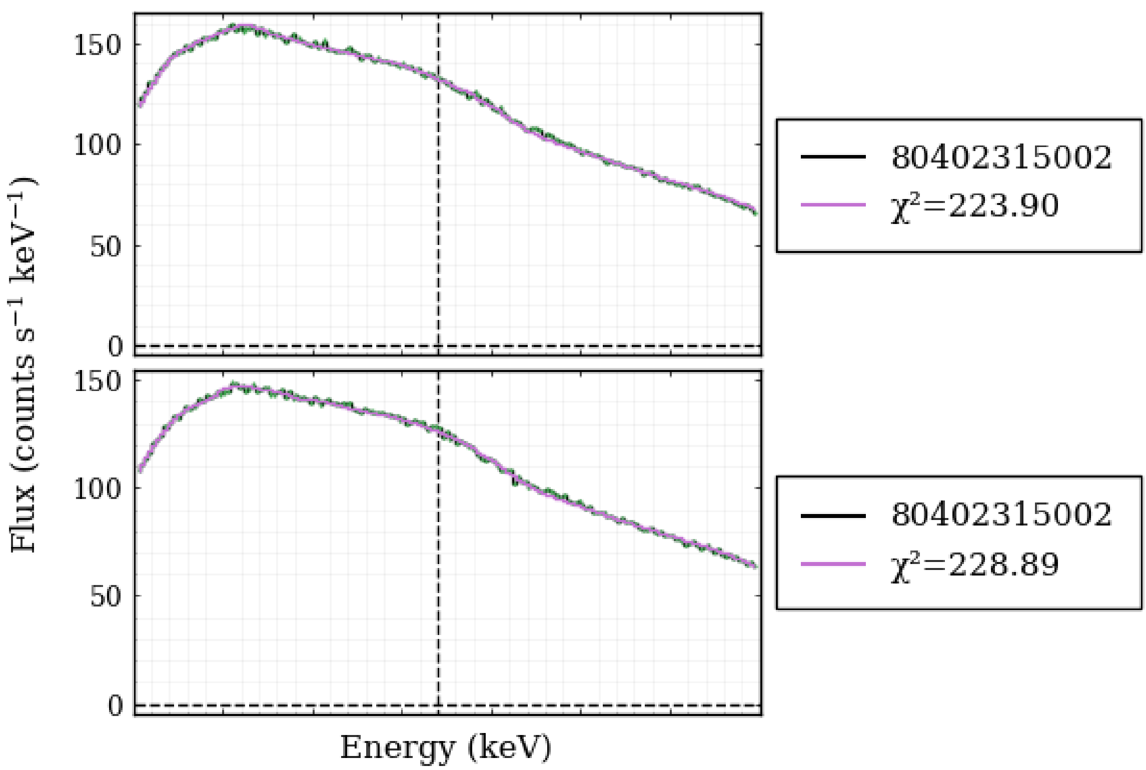


Figure 5.3: The composite fit using the Johannsen metric.

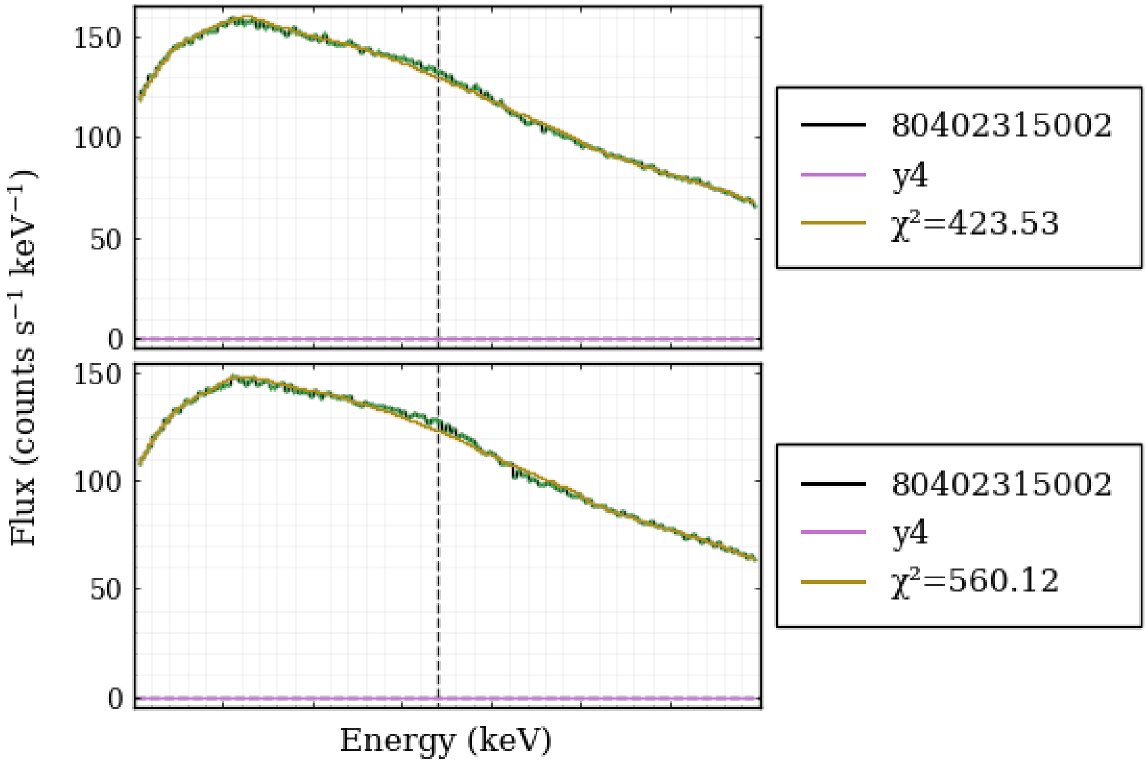


Figure 5.4: The composite fit using the Kerr metric

The inner and outer radii of the accretion disk were set constant at the ISCO and 400 R_g respectively. The Johannsen metric currently seems to yield a better fit, however this appears to be an effect of parameters other than the deformation parameters α_{13} and ϵ_3 as they were both relatively small (1.0285 and 1.6702×10^{-5}, respectively). It may therefore be more effective to investigate the effect of these parameters by first fitting the Kerr metric to the data to find the Kerr parameters, then fitting the Johannsen metric with the found parameters, only allowing the two deformation parameters to vary.

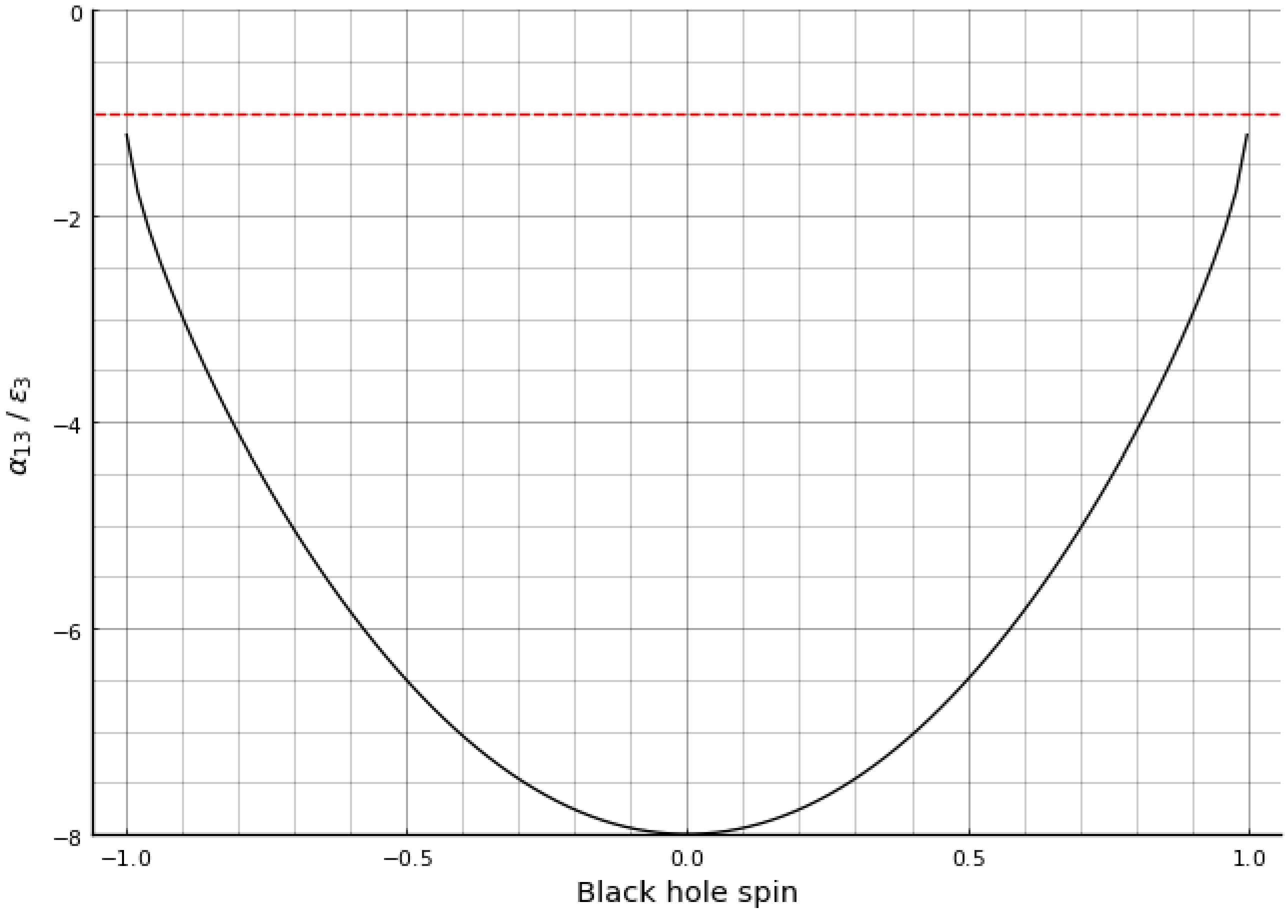
Further reading suggests that it may be possible to constrain the parameters for fitting further. In T. Johannsen's 2013 paper describing the Johannsen metric [3], the following constraints were derived for the deformation parameters:

$$\epsilon_3 > -\frac{(M + \sqrt{M^2 - a^2})^3}{M^3} \tag{5.1}$$

and

$$\alpha_{13} > -\frac{(M + \sqrt{M^2 - a^2})^3}{M^3} \tag{5.2}$$

Until now, the deformation parameters had been allowed to take only positive values. These constraints show that this was potentially a naive restriction, thus the code should be adapted to allow for negative values with these constraints. This would ideally be done through `SpectralFitting.jl`'s parameter patching functionality or equivalent however if that is possible remains to be seen. For now, the lower bound for the deformation parameters has been set to -1, which is allowed no matter how large a gets.



0
−2
−4
−6
−8
α_{13} / ε_3
−1.0
−0.5
0.0
0.5
1.0
Black hole spin

6 Developing and Migrating to a Table Model

6.1 Introduction

Due to having to render the line profile through Gradus upon every iteration of the fit, the model took a significant amount of time to fit to the data. Thus other methods were looked into and it was decided to develop a table model, allowing for the pre-computation of the line profiles. The computed profiles could then be interpolated between for a given set of parameters, significantly speeding up computation. This would of course decrease the accuracy of the model, however the increased efficiency would likely be worth that loss.

The structure for XSPEC table models is well documented at XSPEC Table Models [4], thus the model was constructed using this specification. `SpectralFitting.jl` also has compatibility with this model format, which made integration more viable (see here for more [5]).

6.2 Data Generation

The spectra for the model were simply generated by looping through a defined parameter space, with a given number of values for each of the 5 parameters. The values selected were defined as follows

Parameter	Lower Bound	Upper Bound	Number
a	-0.998	0.998	7
h	1.5	30	10
θ	5	85	10
α_{13}	0	50	7
ϵ_3	0	30	10

This gave a total of 49,000 combinations to loop through. Seven values were used for both a and α_{13} as they had the smallest, and most predictable effect on the line profile, respectively. In a later iteration of this model it would be ideal to have a larger parameter space to reduce the innacuracies of interpolation, however for a test case it was necessary to keep compute times down. The line profiles were stored in columns of a CSV with the header containing the parameter information.

Certain parameter combinations would fail to compute, and therefore a log file was kept, noting the combination and when it failed. This allowed the computations to run over night without risk of an error. The final compute time for this data generation was 16 hours. 3,500 of the 49,000 did not compute correctly, thus there is 45,500 datapoints in the grid.

6.3 Loading into FITS Format

This data was then loaded into a FITS file following the XSPEC model format using the python wrapper for the `heasp` C library, `heasoftpy` (see the documentation for more information). The model specification requires that the grid be rectilinear [4], thus it was not possible to simply disregard the failed runs, and the spectra for these combinations was set to be zeroed out. This was determined to be acceptable as the majority of the failed runs had large α_{13} or ϵ_3, which from earlier testing was noted to be unlikely during fitting, thus the failed combinations would be unlikely to be used for interpolation.

6.4 The Model

To create a model in SpectralFitting using the table, the data was loaded into a `TableModelData` struct. This could then be wrapped using `TableModelInterpolation` such that it can work with the `TableModelInterpolation` function. This functionality made it trivial to set up the interpolation through the parameter grid by simply calling

```
SpectralFitting.interpolate_table!(table, a, h,  ,  13,  3)
```

This returned a spectrum with 1000 datapoints, which could then be interpolated over the domain of the input data. With this architecture it was simple to fit the data as previous.

6.4.1 Performance

Using the nu80402315002 dataset as before, the new model yielded the following results.

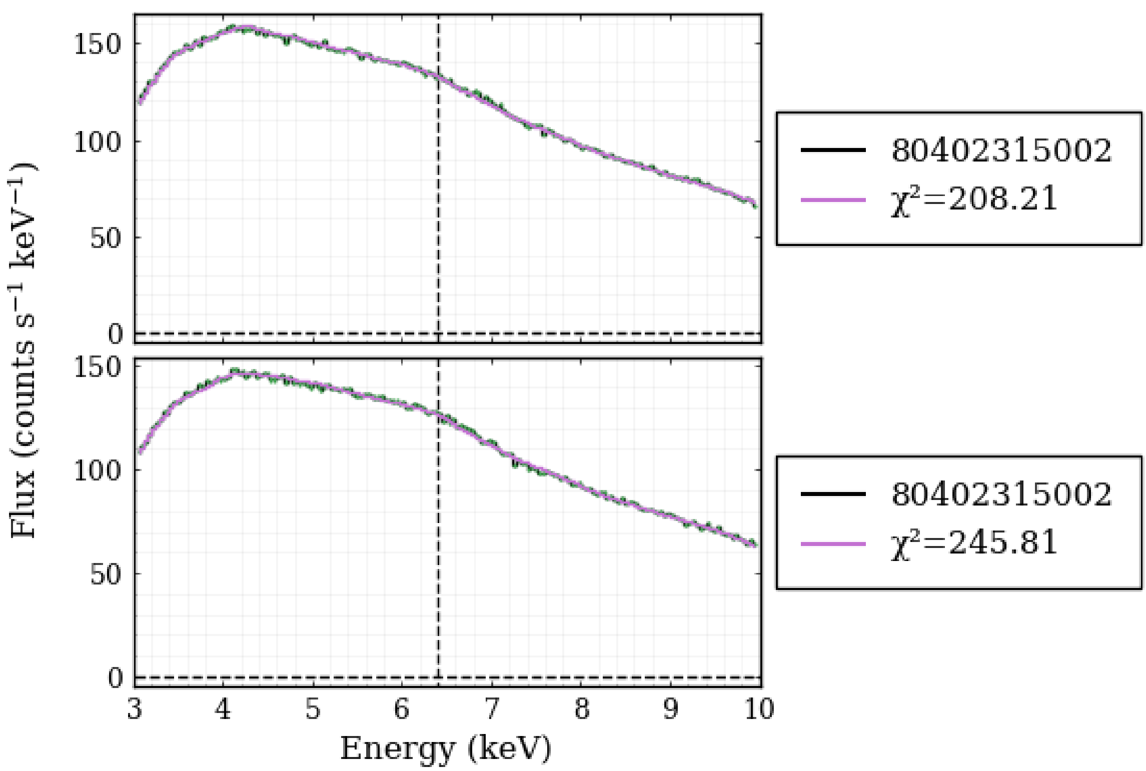


Figure 6.1: The table model result for the Kerr metric

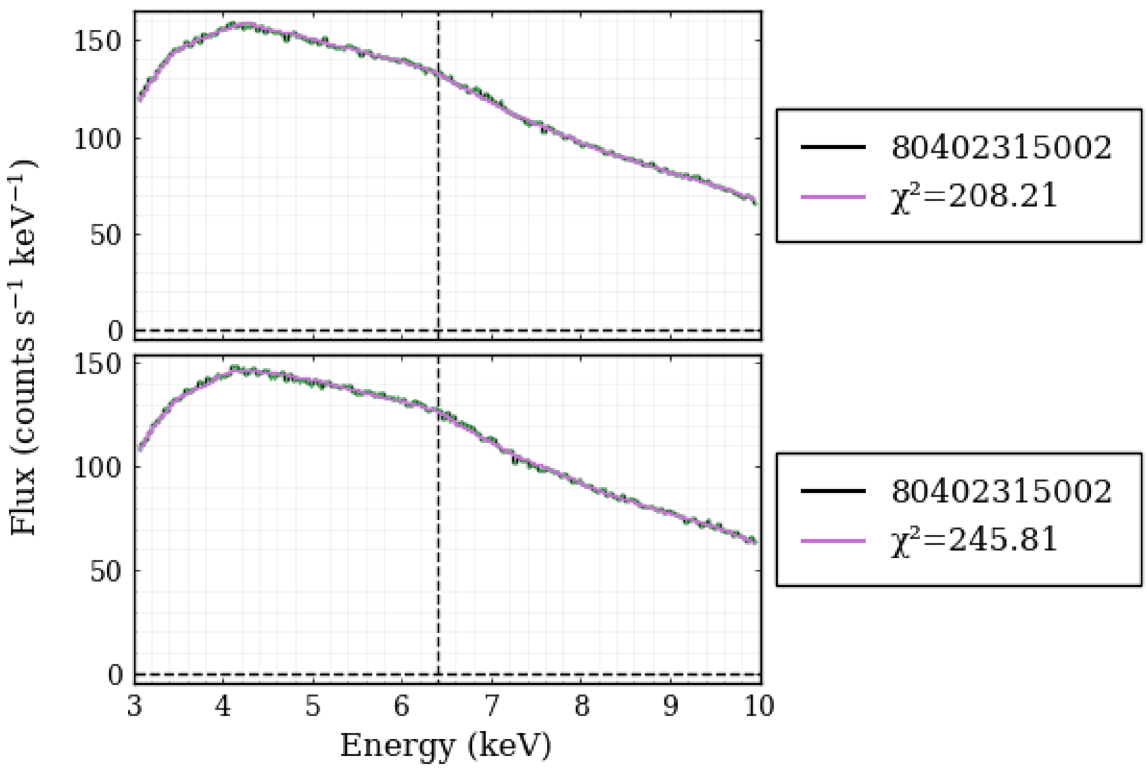


Figure 6.2: The table model result for the Johannsen metric

Comparing to the previous fits, the χ^2 value for both of these fits were surprisingly improved. The Kerr fit was completed by freezing the deformation parameters at 0.

These fits took only 30 seconds to fit and plot both metrics, which was significantly faster than previous which could take an hour or more for a single fit.

6.5 Next Steps

After review of the log file, it was found that a lot of the cases where the deformation parameters were zero failed. This is significant as it implies that many of the Kerr model interpolations are innacurate due to the "correction" method used (setting missing values to zero). Using code to parse the unique values for each of the 5 parameters, it was found that every instance of failure had $h = 1.5$. It was thus decided that upon the next iteration of the model, the lower bound of h would be set to 3. This value was determined after some brief testing with the failed runs, manually setting $h = 3$. $h = 2$ was also attempted, however this similarly caused errors.

Building on what was discussed in the previous chapter regarding the lower bounds for the parameters, it was also tested whether a value of -1 computed correctly for both of the deformation parameters. This was successful and thus a partial implementation of these bounds can be completed.

Further testing was also conducted in the parameter space of a where it was found that the line profile does not change with the sign of a. This will allow for much more resolution without increasing the length of computation.

To use the negative deformation parameters, their allowed values must be reduced as it is necessary to have confidence in the deformation $= 0$ (Kerr) cases. They will therefore take the values $\alpha_{13}, \epsilon_3 \in [-1, 10], \mathbb{Z}$. The other variables will take $h \in [3, 15]$, $a \in [0, 0.998]$ and $\theta \in [5, 85]$. Thus 7 values will be taken for a, 12 for the deformation parameters, and 9 for h, θ, giving a total size of 81648 for the table.

The final iteration that needs to be made upon the data is to increase the domain, as the model is currently forced to extrapolate causing unexpected issues, likely forcing the fitting to take worse values to avoid these errors.

All of these iterations were implemented and another dataset computed. At the time of writing these computations are still ongoing and therefore time was taken to clean up the plotting and do further reading.

Functionality was then added to plot a contour plot of the fit statistic (χ^2) for a pair of fit parameters. The fits are however not currently accurate enough to determine the functionality of this plot, thus this remains to be used.

6.6 Current Issues 2026/07/03

The table model currently does not contain the full set of spectra for negative values of the deformation parameters. For high spin parameters, the code will sometimes fail to compute the line profile citing either one of two errors:

```
ErrorException("No boundaries for minimization could be determined. It is likely this configuration
```

or

```
DomainError([VALUE], "sqrt was called with a negative real argument but will only return a complex r
```

This occurs when attempting to use negative values below ~ -0.4 for either, or both, of the deformation parameters. This is likely either a bug, or an artefact of the Johannsen metric. The only currently known constraints on allowed values of α_{13} and ϵ_3 are those stated in Equations Equation 5.1 and Equation 5.2, which all values used satisfy. The current working theory is thus that this may be a bug with `Gradus.jl` however this requires further research.

There also exists the restriction that $|a| < 1$ to avoid a naked singularity [6], however this is once again satisfied by the used values.

The table model is currently capable of extrapolating, which provides reasonable estimates of the line profile for negative values (see Figure 6.3) however it would not be preferable to rely on extrapolation to constrain the parameters.

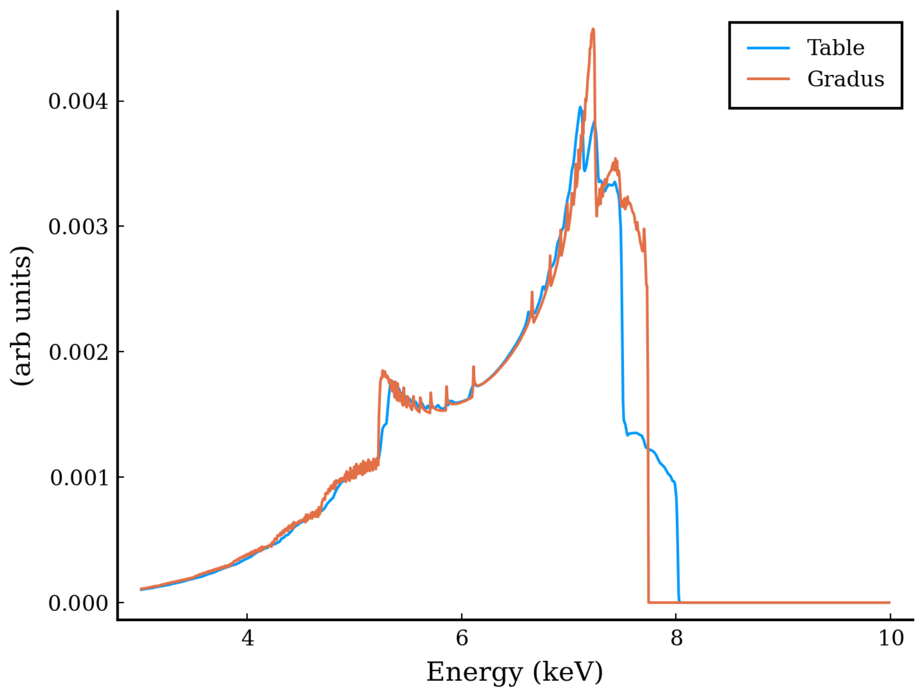


Table
Gradus
0.004
0.003
0.002
0.001
0.000
(arb units)
4
6
8
10
Energy (keV)

Figure 6.3

7 XRISM Data Reduction

These notes are significantly condensed and adapted from the XRISM Quick Start Guide [7]. All key steps for the reduction are on this page, however some details and steps to deal with specific issues have been omitted for brevity. I have noted where applicable where to go for more information.

! Notes

- All commands below assume that HEASOFT and CalDB are installed and instantiated, and that the CalDB installation has the necessary files for XRISM data.
- Data will have observation ID as top directory, e.g. 000125000/
- Pipeline scripts must be run in a directory that contains this directory
- The XRISM Data Analysis Guide contains more information relevant to this process.

7.1 Pipeline Scripts

For all following commands replace `000125000` with the observation id for your data. The commands in this section have been streamlined with InitialPipeline.sh for `bash` systems.

7.1.1 Reprocessing:

In the directory above data directory run the following for the resolve data

```
# Run the pipeline
xapipeline indir=000125000 outdir=000125000_rsl_reproc steminputs=xa000125000 stemoutputs=DEFAULT en
```

Then for the Xtend data

```
# Run the pipeline
xapipeline indir=000125000 outdir=000125000_xtd_reproc steminputs=xa000125000 stemoutputs=DEFAULT en
```

The `.hk` files can be removed as they are duplicates of the corresponding files in the original data. The remaining files can be gzipped for space efficiency. `InitialPipeline.sh` does this automatically.

7.1.2 Analysis Setup

Then make a directory for the analysis and copy the following files

```
mkdir analysis

cp 000125000_xtd_reproc/*_cl.evt.gz analysis/
cp 000125000_rsl_reproc/*_cl.evt.gz analysis/
cp 000125000_rsl_reproc/xa000125000.ehk.gz 000125000_rsl_reproc/xa000125000rsl_px1000_exp.gti.gz 000
```

You will also need the `RA_NOM`, `PA_NOM`, and `DEC_NOM` values from the Resolve header which can be obtained with

```
# Getting Resolve RA, DEC, PA and storing in a file
ftlist xa"$OBSID"rsl_p0px1000_cl.evt.gz K | grep -m 3 -e DEC_NOM -e RA_NOM -e PA_NOM > rsl_coordinat
```

This will write the three values to a text file

7.1.3 PSP Limit Checks

Run PSP limit check to see if the unscreened count rate for Resolve exceeds the psp limit in the `000125000_rsl_reproc/` directory.

```
rslbratios infile=xa000125000rsl_p0px1000_uf.evt.gz filetype=uf outroot=rsl000125000_ufbr
```

This gives a info on if quadrants exceed 50 cts s^-1, which makes the data unusable. A log file is produced which lists the pixels that exceed the limit, these only apply to Resolve data, and these pixels will be unusable.

Next run rslbratios on the cleaned resolve data in the `analysis/` directory:

```
rslbratios infile=xa000125000rsl_p0px1000_cl.evt.gz filetype=cl outroot=rsl000125000_clbr lcbin=128.
```

This gives the same info but for the cleaned data and outputs the following files which will be used later

- `rsl000125000_clbr_2keVto12keV_hist.fits`: Observed vs. predicted branching ratios in the energy range specified by the "eband" parameter
- `rsl000125000_clbr_full_lc.fits`: Full array light curves by grade
- `rsl000125000_clbr_pix_lc.fits`: Per-pixel light curves by grade
- `rsl000125000_clbr_gradeFrac_lc.fits`: Grade fractions versus time
- `rsl000125000_clbr_2keVto12keV_pixMap.img`: 20 images of counts by grade, and grade fractions for two energy bands

7.2 Removing Anomalous Pixels from Xtend Data

> **! Important**
>
> All following commands are run from the `analysis/` directory

> **i Note**
>
> This section makes heavy use of `xtdpixclip`, and the usage is not documented in the XRISM QSG, see the `xtdpixclip` documentation [8] for more information.

First generate region files using ds9, viewing the image in DET coordinates with the command.

```
ds9 "xa000125000xtd_p031100010_cl.evt.gz[bin=detx,dety]"
```

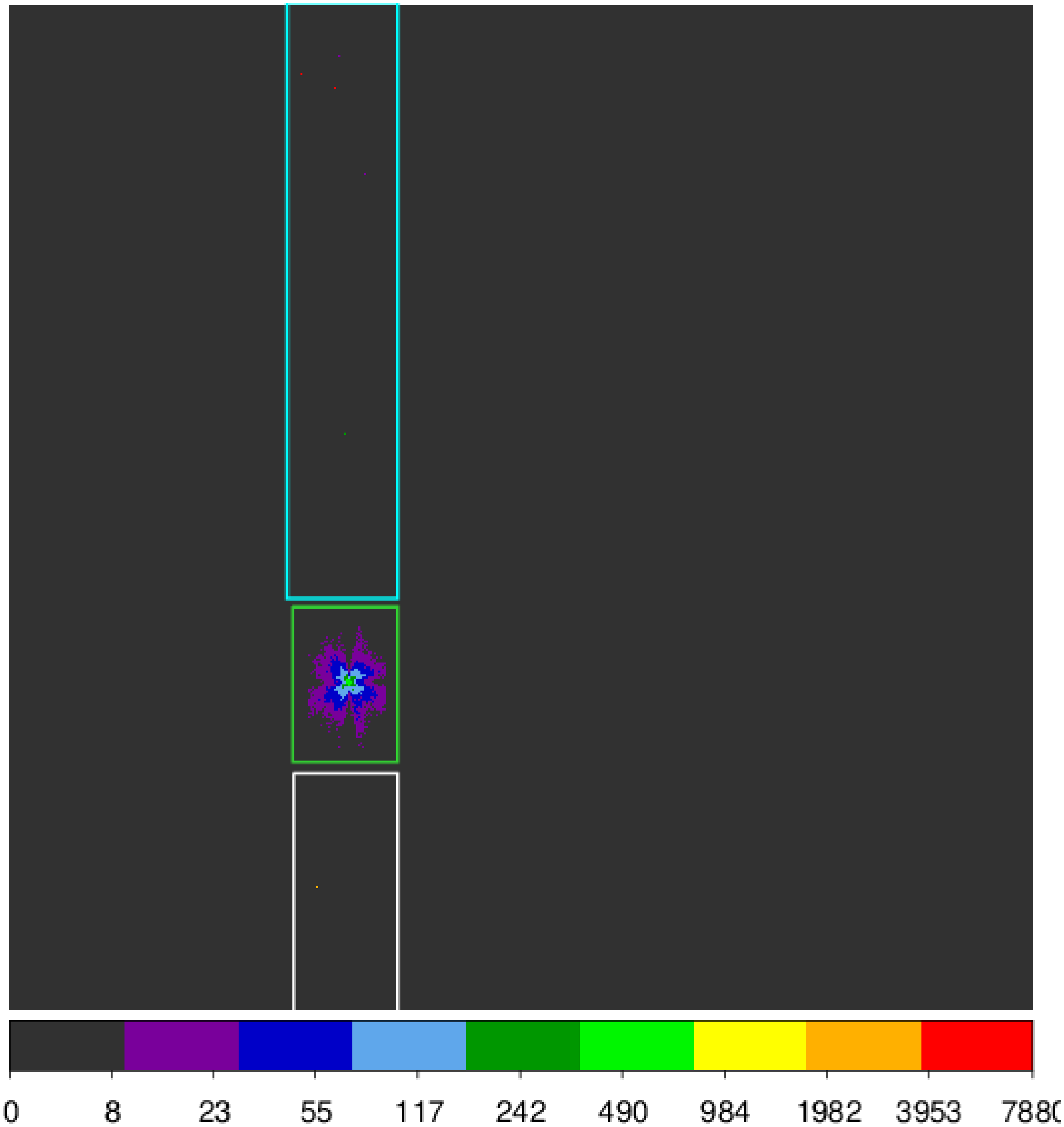


Figure 7.1: The regions used.

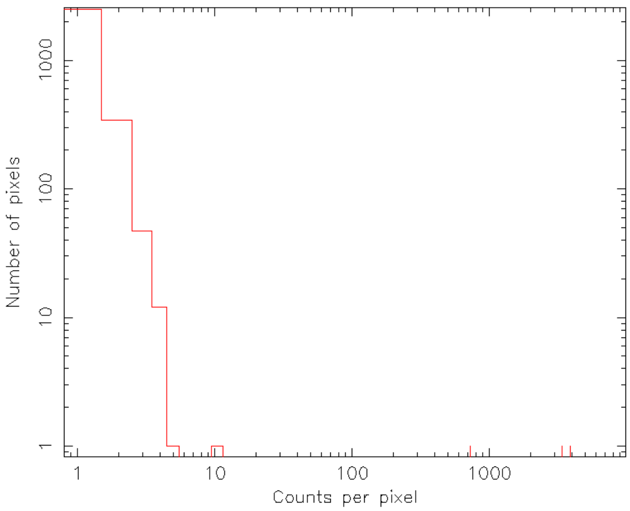
Three regions were selected and labelled as follows:

- White: `1_bottom.reg`
- Green: `1_src.reg`
- Cyan: `1_top.reg`

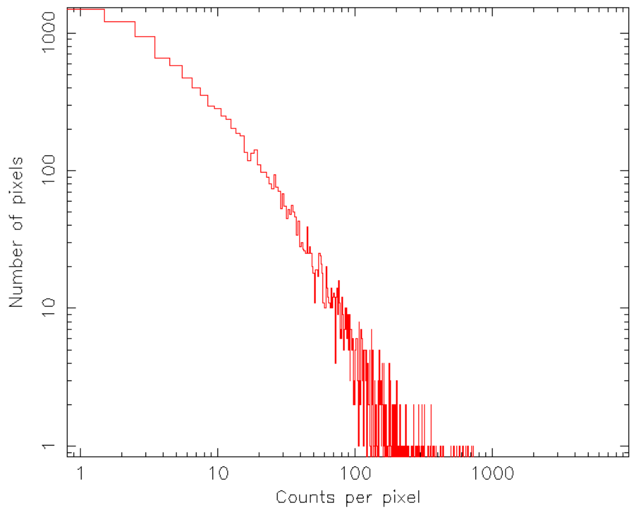
Next run xtdpixclip in histogram mode to generate histograms for the regions, including all of the region files just generated.

```
punlearn xtdpixclip
xtdpixclip evtinfile="xa000125000xtd_p031100010_cl.evt.gz" outroot="xa000125000xtd_p031100010_cl" pr
```

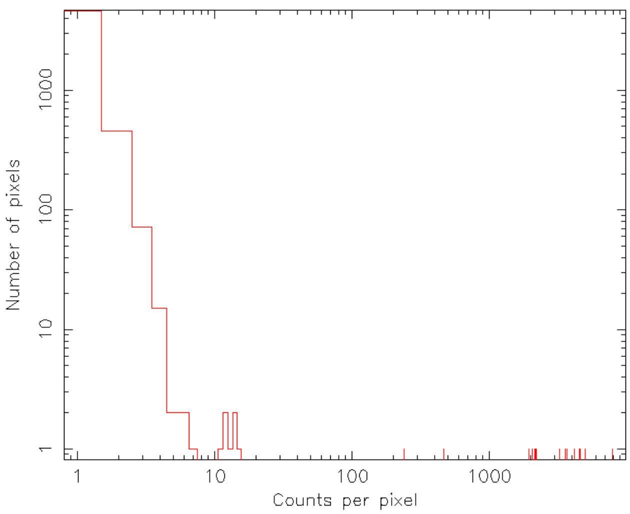
These histograms can be used to determine the thresholds to use as cutoffs for anomolous pixels. The output histograms are shown below



(a) The bottom region histogram



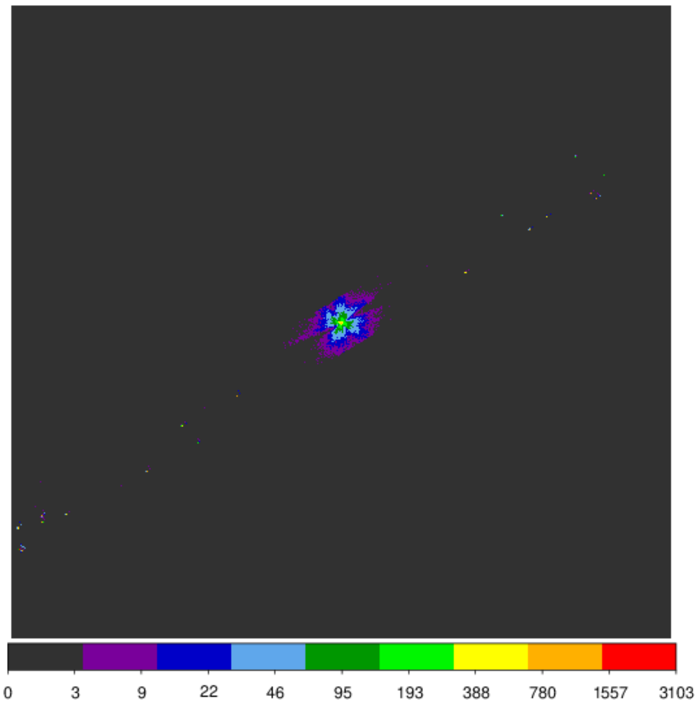
(a) The source region histogram



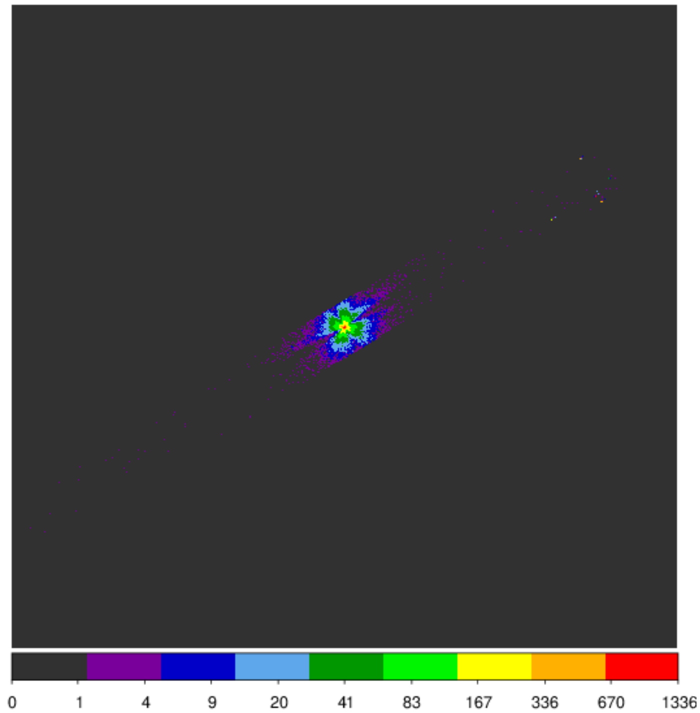
(a) The top region histogram

For the two non-source regions a threshold of 8 was selected, and for the source region 1000 was selected. With the regions selected run xtdpixclip again but in apply mode, entering the thresholds selected

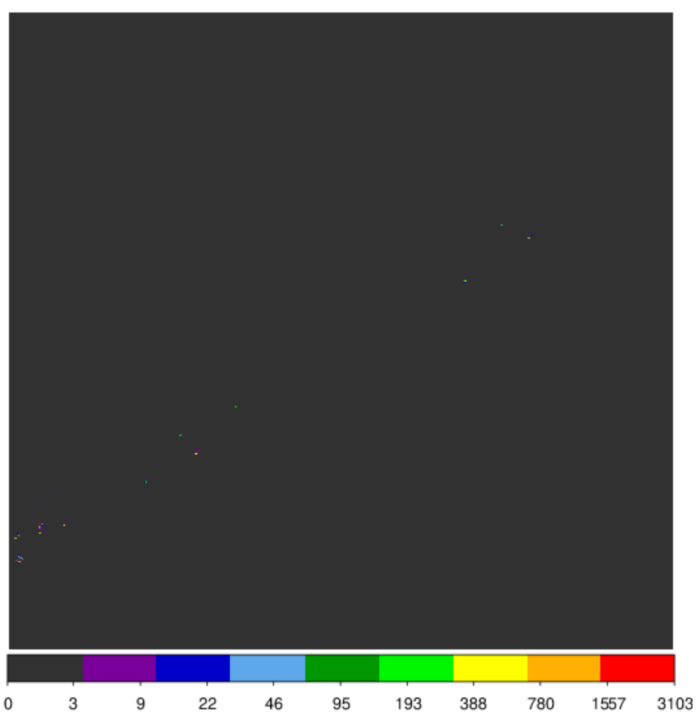
```
punlearn xtdpixclip
xtdpixclip evtinfile="xa000125000xtd_p031100010_cl.evt.gz" outroot="xa000125000xtd_p031100010_cl" pr
```



(a) The Raw image



(a) The cleaned image



(a) The filtered out pixels

7.3 Proximity and Rise time screening

- Proximity screening: Removing frame events based on the unlikelihood of events happening too close together in a given time interval. This must be balanced against the loss of counts.

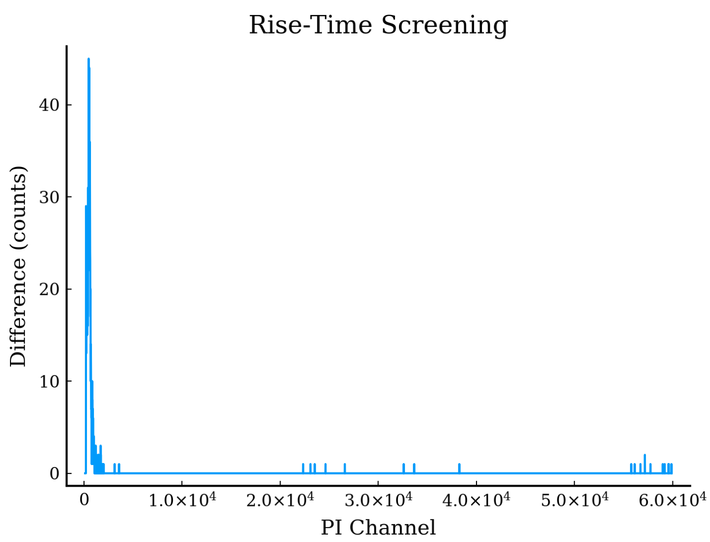
The effect of both can be previewed through the files:

```
rsl000125000_clbr_prox_Hp_spectra.fits
rsl000125000_clbr_rise_Hp_spectra.fits
```

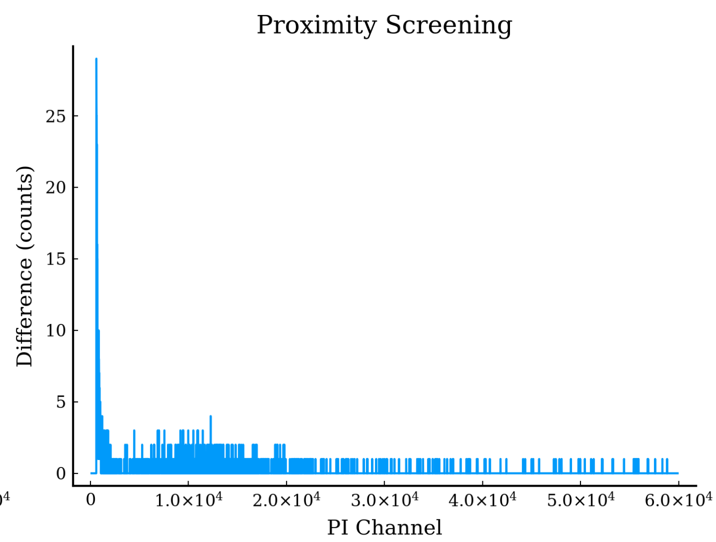
These can be examined by plotting the difference after proximity and rise-screening to observe the losses. This can be done with fplot:

```
# For Proximity Screening
fplot rsl000125000_clbr_prox_Hp_spectra.fits[DIFFERENCE] PI DIFF - /PS Q

# For Rise-Time Screening
fplot rsl000125000_clbr_rise_Hp_spectra.fits[DIFFERENCE] PI DIFF - /PS Q
```



(a) The Rise-time filtering difference



(a) The Proximity filtering difference

The losses for each were determined to be acceptable for this dataset, thus the both filters were applied with the command:

```
ftcopy infile="xa000125000rsl_p0px1000_cl.evt[EVENTS][(PI>=600) && (((((RISE_TIME+0.00075*DERIV_MAX)
```

To skip proximity filtering you can omit `&&STATUS[4]==b0`.

> **Note**
>
> The QSG includes a section here on the `xaspecratios` tool which is not available until HEASoft 6.37 and is therefore omitted from these notes.

7.4 Extract Broadband Images and Establish Source Center Coordinates

- Check the value of the EXPOSURE keyword in the cleaned event files, if the exposure time is unexpectedly low then consult section 9 of the QSG [7].
- You should now have the following files
  - `xa000125000rsl_p0px1000_cl2.evt` (Resolve) and
  - `xa000125000xtd_p031100010_cl.evt` or `xa000125000xtd_p031100010_xpc_clnevt.fits` (Xtend)

Start XSELECT and make a 2-10 keV resolve image in DET coordinates as:

```
read eve xa000125000rsl_p0px1000_cl2.evt .
set image DET
filter pha_cutoff 4000 20000
extr image
save image xa000125000rsl_p0px1000_detimg.fits
```

Observing this image in `ds9` or otherwise, *the centre of the source in the Resolve data should be noted down.* This should be close to (3.5, 3.5), however we only need to estimate this approximately as it is used as a sanity check against the RA and DEC coordinates. For most standard observations we can simply assume this centre coordinate.

Next copy over a resolve region file from the `$HEADAS` directory.

```
cp $HEADAS/refdata/region_RSL_det.reg .
```

and make another region file `exclude_calsource.reg` containing

```
physical
-circle(920.0,1530.0,92.0)
-circle(919.0,271.0,91.0)
```

using

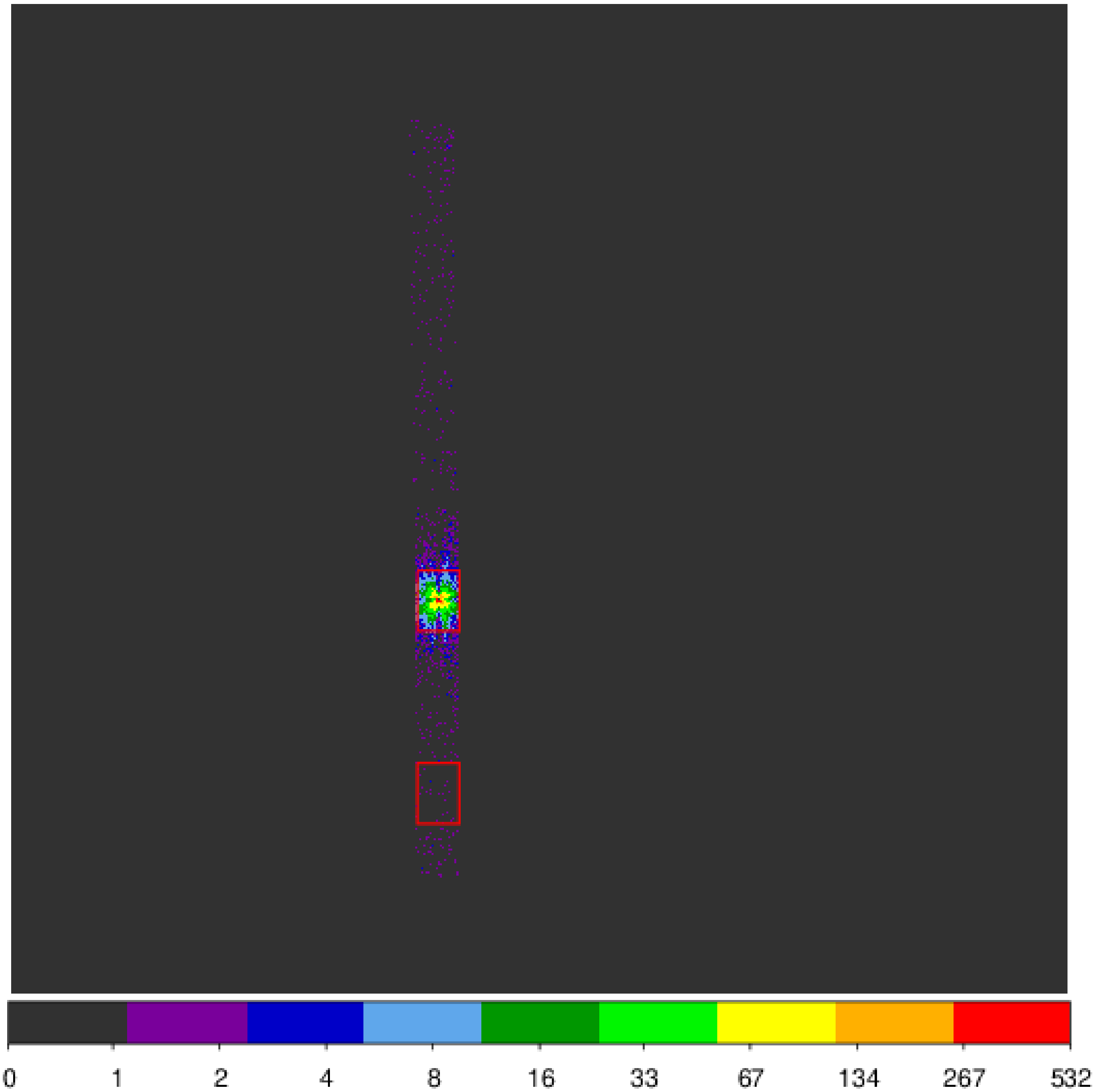
```
echo "physical" > exclude_calsources.reg
echo "-circle(920.0,1530.0,92.0)" >> exclude_calsources.reg
echo "-circle(919.0,271.0,91.0)" >> exclude_calsources.reg
```

Next make a 0.5-10 keV Xtend image in DET coordinates

```
clear all
read eve xa000125000xtd_p031100010_cl_xpc_clnevt.fits .
set image DET
filter region exclude_calsources.reg
filter pha_cutoff 83 1667
extr image
save image xa000125000xtd_p031100010_detimg.fits
```

Now we can estimate the centroid of the source in DET coordinates in the Xtend data using `ds9`. *Note this coordinate down as it will also be useful later.*

Next make region files for the source and background lightcurves and spectra. It is reccommended to use a rectangle of length 5 arcmin and width as wide as possible for 1/8 Window mode images. The region should not extend beyond the chip boundary as this will result in an incorrect ARF. For full window mode it is reccommended to use a circle of radius 2.5 arcmin. The regions selected for the data used here are shown below



and were saved as

```
region_000125000xtd_src.reg
region_000125000xtd_bgd.reg
```

7.5 Extract Lightcurves and Spectra

In the analysis/ directory start a new XSELECT session and read in the cleaned resolve event file:

```
read eve xa000125000rsl_p0px1000_cl2.evt .

# Extract a resolve full-array lightcurve in the 2-10 keV band (time bin in seconds)
set image det
filter pha_cutoff 4000 20000
set binsize 128.0
extr curve exposure=0.8
save curve xa000125000rsl_allpix_b128_lc.fits

# If a time bin has less than a fraction "exposure" then you will get a warning about too many bins

# Extract a Resolve spectrum from the full array, for hi-res primary events only
# The first command can be modified to exclude pixels or a group of pixels
filter column "pixel=0:11,13:35"
filter GRADE "0:0"
extr spectrum
save spectrum xa000125000rsl_allpix_Hp_src.pi

# Read in the cleaned Xtend event file and extract 0.5-10 keV lightcurves and spectra from the backg
clear all
read eve xa000125000xtd_p031100010_cl_xpc_clnevt.fits .
set image det
filter region region_000125000xtd_bgd.reg
filter pha_cutoff 83 1667
set binsize 128.0
extr curve exposure=0.4
save curve xa000125000xtd_0p5to10keV_b128_bgd_lc.fits
clear pha_cutoff
extr spectrum
save spectrum xa000125000xtd_bgd.pi
clear region
filter region region_000125000xtd_src.reg
filter pha_cutoff 83 1667
extr curve exposure=0.4
save curve xa000125000xtd_0p5to10keV_b128_src_lc.fits
clear pha_cutoff
extr spectrum
save spectrum xa000125000xtd_src.pi
exit
```

7.6 Check Attitude Stability

```
ftselect infile=xa000125000.ehk outfile=xa000125000_gtifiltered.ehk
# Using the expression:
gtifilter("xa000125000rsl_p0px1000_cl2.evt[GTI]")

fplot xa000125000_gtifiltered.ehk TIME ANG_DIST - /PS Q outfile=attitudestability.ps
```

Examine the attitude stability during the observation (e.g. by plotting ANG_DIST versus TIME for the "xa000125000_gtifiltered.ehk" file)

If you need to extract products with stricter attitude stability, make an appropriate GTI file. You can also make ARFs that are based on more than one attitude bin. Read the help for xaexpmap and xaarfgen for guidance on how to do that.

7.7 Make Exposure Map/Attitude Histogram Files

The following examples make histograms with one attitude bin, which should be good enough for the majority of cases.

> ⚠ Mitigation of a bug in CalDB 12 and HEASoft 6.36
>
> If you are using CalDB 12 and HEASoft 6.36, there is a bug related to new optical axes offsets that affect the ARF calculation, which will be fixed in subsequent CalDB and software releases. Until then, a mitigation strategy is applied, as shown in commands below, that reverts creation of the ARF to be the same as if the previous CalDB and software versions were used.

For Resolve:

```
punlearn xaexpmap

xaexpmap ehkfile=xa000125000.ehk.gz gtifile=xa000125000rsl_p0px1000_cl2.evt instrume=RESOLVE badimgf
```

For CalDB 12 and HEASoft 6.36 mitigation:

```
fparkey fitsfile=xa000125000rsl_p0px1000.expo keyword=OPTAOFFX value=0.0 comm="optical axis X offset
fparkey fitsfile=xa000125000rsl_p0px1000.expo keyword=OPTAOFFY value=0.0 comm="optical axis Y offset
```

For Xtend:

```
xaexpmap ehkfile=xa000125000.ehk.gz gtifile=xa000125000xtd_p031100010_cl_xpc_clnevt.fits instrume=X
```

For CalDB 12 and HEASoft 6.36:

```
fparkey fitsfile=xa000125000xtd_a031100010.expo keyword=OPTAOFFX value=0.0 comm="optical axis X offs
fparkey fitsfile=xa000125000xtd_a031100010.expo keyword=OPTAOFFY value=0.0 comm="optical axis Y offs
```

7.8 Making Spectral Response Files

Two methods are listed in the QSG, for these notes I used the first.

7.8.1 Checking Coordinates

We first check the coordinates through `coordpnt` for each instrument. Replace `"RDETX0, RDETY0"`, and `"XDETX0, XDETY0"` with the coordinates relevant to the Resolve and Xtend data noted down earlier. `RA_NOM`, `DEC_NOM`, and `PA_NOM` can be taken from the `rsl_coordinates.txt` file generated earlier.

```
punlearn coordpnt
coordpnt input="RDETX0,RDETY0" outfile=NONE telescop=XRISM instrume=RESOLVE teldeffile=CALDB startsy

punlearn coordpnt
coordpnt input="XDETX0,XDETY0" outfile=NONE telescop=XRISM instrume=XTEND teldeffile=CALDB startsys=
```

The output from these commands are in degrees and should be compared against a trustworthy catalogue for the object you are looking at. If they are within 10" or so of the expected coordinates then continue to the next step, otherwise use the outputs from these commands to generate the ARFs with `xaarfgen` (see later).

The true coordinates were taken from NASA as (12:10:32.6,+39:24:21) (RA,DEC) and the output from both commands above were within 10" of this.

Before running xaarfgen or rslmkrsp, in order to avoid version conflicts, delete the parameter files for `xaarfgen` and `xaxmaarfgen` in your local `pfiles/` directory. For example:

```
rm $HOME/pfiles/xaarfgen.par
rm $HOME/pfiles/xaxmaarfgen.par
```

7.8.2 Make RMFs

For the two Resolve commands in this section replace `NGC4151` with a label for the object you're using.

We first make a special event file that removes all Low Secondary (Ls) events. We calculate the Hp fraction over a clean energy range for which we choose 3-10 keV to maximise accuracy.

> **Note**
>
> All of the following commands can be automatically run sequentially using `RMFARF.sh`. Note that this will only work for a non-bright point source as the pipeline differs slightly for other sources.

```
ftcopy infile="xa000125000rsl_p0px1000_cl2.evt[EVENTS][(PI>=6000)&&(PI<=20000)&&(ITYPE<4)]" outfile=
```

Make a large full-array Resolve RMF:

```
punlearn rslmkrmf
rslmkrmf infile=xa000125000rsl_p0px1000_UPR.evt outfileroot=NGC4151_obs1_60k_60k_Hp_allpix_L regmode
```

Make a small full-array Resolve RMF:

```
punlearn rslmkrmf
rslmkrmf infile=xa000125000rsl_p0px1000_UPR.evt outfileroot=NGC4151_obs1_60k_60k_Hp_allpix_S regmode
```

Neither of the above includes the electron-loss continuum which may make the source appear to have a soft excess. Section 14.6 of the QSG includes instructions on how to make an extra large Resolve response matrix.

Make an Xtend RMF:

```
punlearn xtdrmf
xtdrmf infile=xa000125000xtd_src.pi outfile=xa000125000xtd_p031100010_src.rmf rmfparam=CALDB eminin=
```

7.8.3 Make ARFs

For Resolve data:

If the resolve data has Hp fractions that vary significantly from pixel to pixel see the QSG to make spectral responses (combined RMF and ARF files). In the following commands replace `RA` and `DEC` with the values determined earlier from `coordpnt`, for the dataset used here 182.63409879 and 39.40629018 were used

> **Note**
>
> If you excluded any pixels from the spectrum then replace `region_RSL_det.reg` with an appropriate region file excluding these pixels.

To make a full-array point source ARF for Resolve, run

```
punlearn xaarfgen
xaarfgen xrtevtfile=raytrace_xa000125000rsl_p0px1000_ptsrc.fits source_ra=RA source_dec=DEC telescop
```

For Xtend:

```
punlearn xaarfgen
xaarfgen xrtevtfile=raytrace_xa000125000xtd_p031100010_boxreg_ptsrc.fits source_ra=RA source_dec=DEC
```

8 Further Constraints on the Deformation Parameters

Following a discussion with my supervisors it was found that the issues with negative deformation parameters were not an issue with Gradus but a quirk of the metric. To explore this, a heatmap was plotted of the ISCO for combinations of α_{13} and ϵ_3

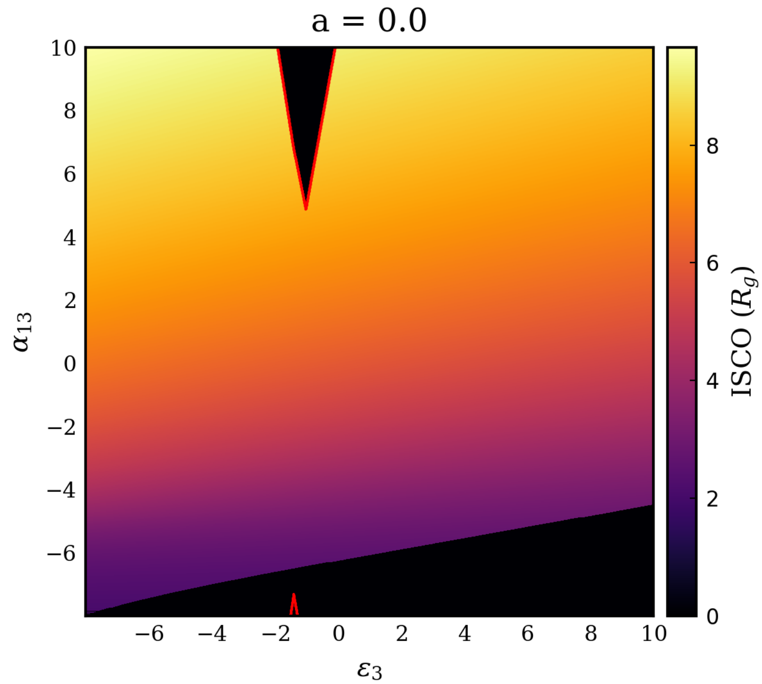


Figure 8.1: The valid deformation parameters for a Schwarszchild black hole. The black regions show invalid configurations, and the red contours denote regions where there is a naked singularity.

Doing several of these plots for different values of the spin parameter may permit rigorous constraints of the allowed values of the deformation parameters. This may be done in two ways:

1. Fitting the data with the Kerr metric to constrain the value of a such that bounds can be determined for the deformation parameters in advance.
2. Somehow dynamically setting constraints on the parameters by selecting from a pre-computed table of allowed combinations for all values of a (permitted by interpolation).
3. Checking the configuration upon each iteration of the fit, and either rejecting the configuration, or manually setting it to an acceptable configuration if it lies within an unphysical region.

The first method would allow for an exploration as to whether the deformation parameters can independently improve the fit of the model, but may reduce the accuracy of the overall fit through forbidding the spin parameter to change. This may be overcome by using the method to obtain a "best guess" for the parameters and then using the result of this for a third and final fit, however this may require implementing the second method also.

The second method would likely be more computationally complex and introduce an increased amount of uncertainty due to the reliance on interpolation.

The third method is potentially the least complex and introduces the least error as it allows the fit to freely explore the parameter space while guiding it through the process to ensure the results remain physical.

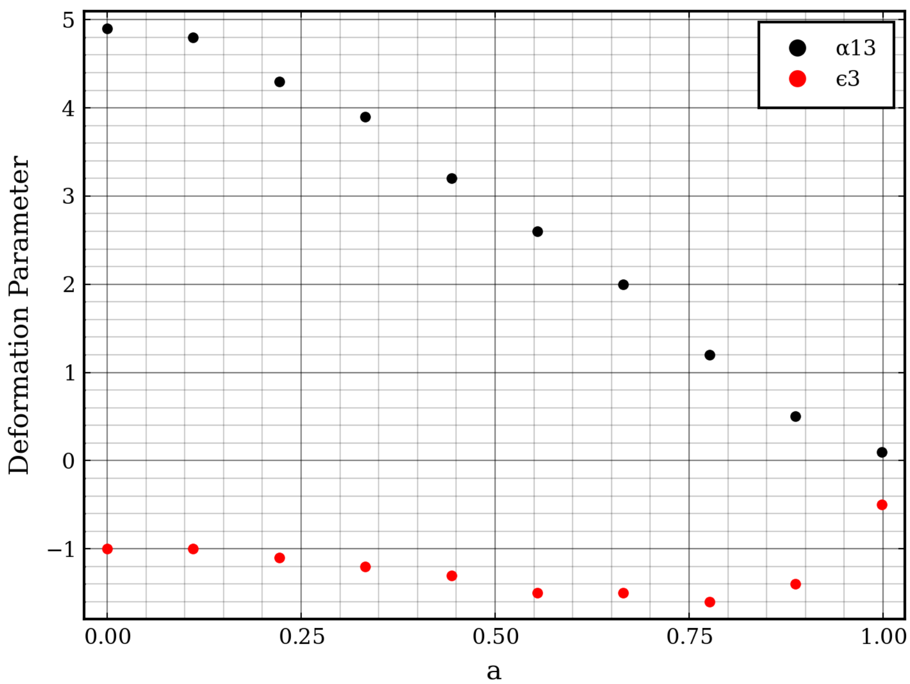
8.1 The Behaviour of the Unphysical Regions

To gather more data on the allowed parameter space, the above image was replicated for 5 spin values as shown below:

From these plots it seems that it may be possible to empirically parameterise the invalid regions with 5 linear equations: one for the wedge in the bottom right first 3 figures, and two for each of the triangles visible in all cases.

It may be practical to run a selection of cases to find the vertices of these regions such that their position can be plotted as a function of a and then fit to find an empirical expression for the allowed values for any a.

To test this hypothesis a quick plot was made with 10 samples of the lower vertex of the triangular region seen in all plots at the top. The result of this is shown below

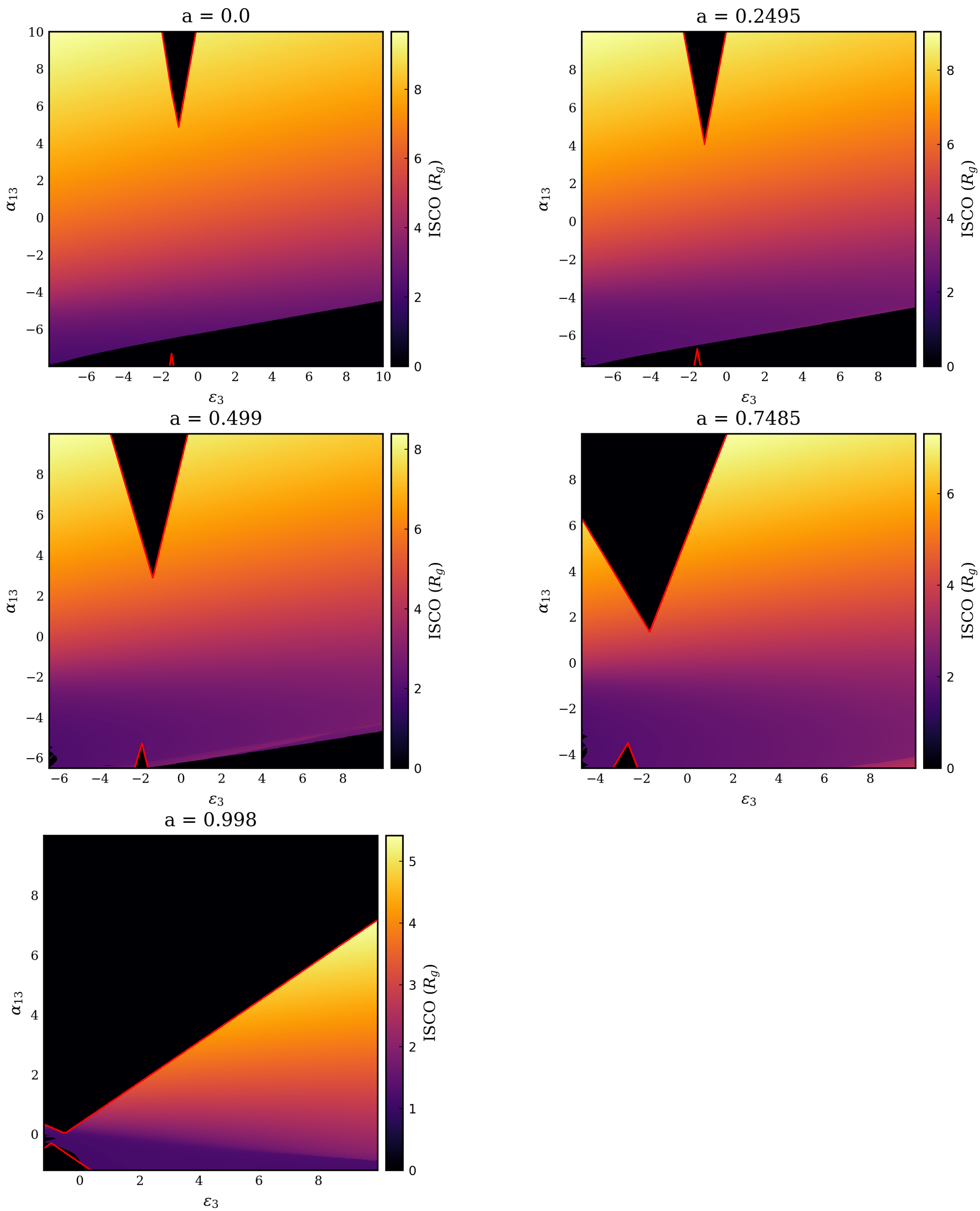


These example coordinates were found manually inspecting the heatmap with plotly. To find them more precisely, several high definition heatmaps could be produced and the resulting matrix iterated through to find the extrema of each of the shapes. The parameter space could then be constrained by either interpolating the values within a table, or by fitting some function to the data and finding the vertices of these major shapes.

The vertices can then be used to define bounded exclusion regions, within which the model is invalid. During fitting these exclusion regions can then be used to “reject” a fit and force the parameters into an acceptable range.

8.2 Allowed Values of the Deformation Parameters for Varying Spin

As another investigative route, Figure 6 from [3] was reproduced as below. It was found that the results held consistent between Gradus, and Johannsen’s original calculations. The α_{13} parameter space showed an additional


a = 0.0
a = 0.2495
a = 0.499
a = 0.7485
a = 0.998
α_{13}
ε_3
ISCO (R_g)

region (shown in red) where a naked singularity is present. This was not considered previously, however removes a significant region from the parameter space and will thus be considered here.

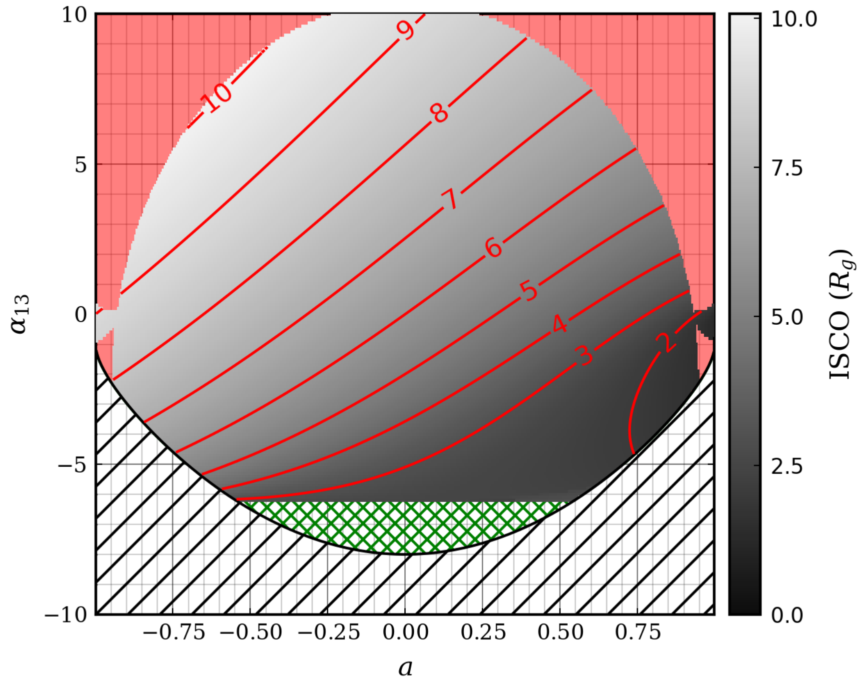


Figure 8.2: The valid values for ϵ_3 to take for a range of spin values. The black hatched region marks the constraints placed by Equation 5.2. Contours of constant ISCO are marked in red. The shaded red region marks where a naked singularity occurs.

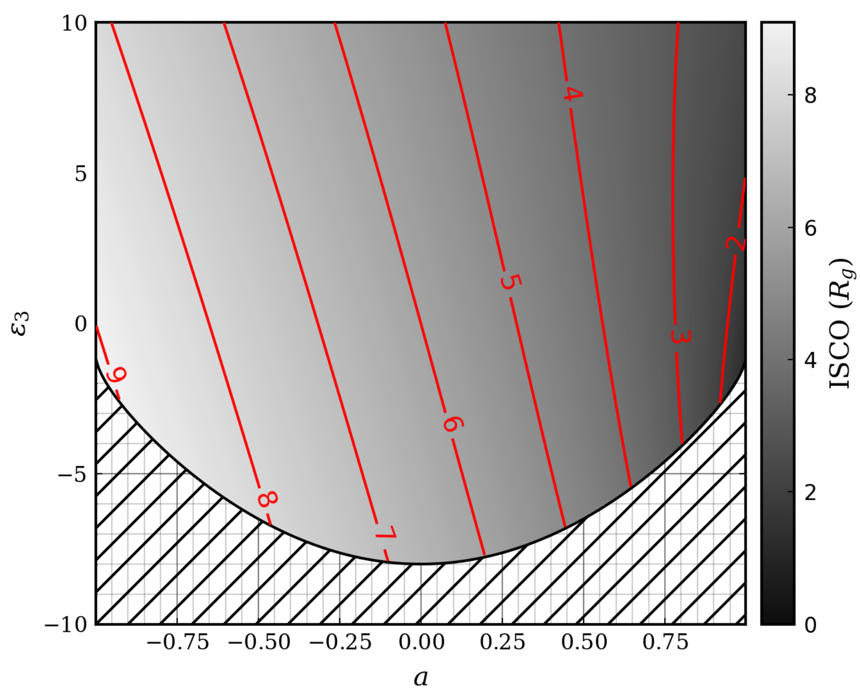


Figure 8.3: The valid values for ϵ_3 to take for a range of spin values. The black hatched region marks the constraints placed by Equation 5.1. Contours of constant ISCO are marked in red.

8.3 Implementation

8.3.1 Plan 2026/07/10

The current plan for excluding these unphysical regions of the parameter space from the fitting is as follows. The unphysical regions will be defined by a set of equations for a given spin, and within these regions the fit will be setup to check if the combination is valid. To ensure no invalid combinations are mistakenly used, some padding will be added to these equations such that all invalid combinations are contained within. When the fitting iterator enters an invalid region, the combination will be explicitly checked for naked singularities, or the absence of a valid ISCO.

If the region is found to be invalid, then the `invoke!` function will return an array of zeros, which will signal to the fitting iterator that the parameter combination is not allowed for fitting.

A potential roadblock to this process is the unpredictable region in the bottom left corner of the high spin case. This will be dealt with by defining a circular region around this area to catch all of these cases. This will lead to a decrease in efficiency, but will increase the accuracy and thoroughness of the model.

8.3.2 Notes Following Implementation

The boundary lines were successfully defined by finding the vertices through a grid search defined in `Code/utils/DeformUtils.jl`, and a csv of gradients and intercepts was defined for quick reading of the lines. Padding was added manually to these lines by randomly sampling the parameter space and checking the validity of the combination both using the bounds and explicitly checking the metric. Whenever the bounds missed an invalid point in the parameter space they were adjusted slightly and the process repeated until 10^6 consecutive random samples could be taken without a point being missed. The bounds were designed to be overzealous such that no valid combinations are missed.

To avoid removing valid combinations, the fitting process will involve first a quick check with the boundaries, followed by a thorough check with the metric if the first check fails.

The "unpredictable region" noted in the lower left corner was bounded with a line rather than a circle for simplicity.

The call to the bounds within the fitting iterator is shown below

```
# Reading from table for a quick check of parameter validity
if !QuickIsValid( 3,  13, a)
    # Doing a finer check if the quick check fails
    if !IsValid( 3,  13, a)
        output = fill(NaN, length(input)-1)
        return output
    end
end
```

Here `QuickIsValid` checks the combination using the defined bounds and `IsValid` checks it thoroughly using the metric through `Gradus.jl`. If both checks find the parameter combination to be invalid then the model returns an array of `NaN` for that iteration, such that the iterator will calculate a bad fit and move out of the region. The conditions that are checked through these functions are:

- The presence of a valid ISCO: Checked by defining a metric through `Gradus.jl` and attempting to calculate the isco using `Gradus.ISCO`
- The presence of a naked singularity: Checked using `Gradus.is_naked_singularity`

- If the combination satisfies Equation 5.2 and Equation 5.1
- If the magnitude of the spin is below 0.998

The final check is done as the fitting iterator sometimes exceeds the upper limit, causing the interpolation for the bound definitions to try to extrapolate which causes errors.

The next step is to create a final table model to use for the fit that covers the entire allowed parameter space. In the construction of this, each combination will be checked for a valid ISCO and if Equation 5.2 and Equation 5.1 are satisfied. If any of these are not satisfied then that point in the table will be an array of `NaN` to ensure that it does not get used.

References

[1] A. C. Fabian *et al.*, Publications of the Astronomical Society of the Pacific **112**, 1145 (2000).
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[6] A. Tripathi *et al.*, Phys. Rev. D **99**, 083001 (2019).
[7] XRISM Science Data Center, *XRISM Quick-Start Guide* (NASA, 2025).
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